

Counting partial Hadamard matrices in the cubic regime

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Abstract

We give a precise asymptotic formula for the number of $n \times 4t$ partial Hadamard matrices in the regimes $t/n^3 \rightarrow \infty$ and $t/n^3 \rightarrow \Theta$ for sufficiently large fixed Θ . This strengthens earlier results of de Launey and Levin, who obtained the asymptotic for $t/n^{12} \rightarrow \infty$, and of Canfield, who extended this to $t/n^4 \rightarrow \infty$.

1 Introduction

The Hadamard conjecture asks whether, for every positive integer n divisible by 4, there exists an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$. Despite sustained effort since Sylvester’s recursive construction [?] and Paley’s finite-field families [?], the conjecture remains open; we refer to Horadam [?] and Seberry–Yamada [?] for background on Hadamard matrices, to Tressler [?] and Browne et al. [?] for surveys, and to Cati–Pasechnik [?] for a recent computational database.

Rather than constructing full Hadamard matrices, one can study partial ones: an $n \times t$ matrix with entries in $\{\pm 1\}$ is a *partial Hadamard matrix* if its rows are pairwise orthogonal. Partial Hadamard matrices of width nearly $2n$ are known to exist for large n by combining constructions from combinatorial design theory [?, ?] with analytic number theory [?] (see [?, Theorem A]). While these results settle existence for widths close to $2n$, a finer question is to determine how many partial Hadamard matrices there are as a function of n and t .

De Launey and Levin [?] introduced a Fourier-analytic framework that answers this question for large t . Their approach encodes the row-orthogonality conditions for an $n \times t$ matrix through the $d = \binom{n}{2}$ pairwise products $y_i y_j$: writing $Z(y) = (y_i y_j)_{1 \leq i < j \leq n} \in \{\pm 1\}^d$, a matrix $Y = [y^{(1)} \cdots y^{(t)}]$ with ± 1 entries is partial Hadamard if and only if $\sum_{r=1}^t Z(y^{(r)}) = 0$; see Figure 1. Counting partial Hadamard matrices therefore reduces to counting returns to the origin of the random walk $S_t = \sum_{r=1}^t Z(\xi^{(r)})$ on $\mathbb{Z}^d$, where $\xi^{(1)}, \xi^{(2)}, \dots$ are i.i.d. uniform on $\{\pm 1\}^n$. If $N_{n,t}$ denotes the number of $n \times t$ partial Hadamard matrices,

$$N_{n,t} = 2^{nt} \mathbb{P}(S_t = 0), \tag{1}$$

and the Fourier inversion formula [?, P3, p. 57] gives

$$\mathbb{P}(S_t = 0) = (2\pi)^{-d} \int_{\mathbb{T}^d} \psi(\lambda)^t d\lambda, \quad \psi(\lambda) = \mathbb{E}[e^{i\lambda \cdot Z(\xi)}]. \tag{2}$$

In particular, the Hadamard conjecture is equivalent to the positivity of the integral (2) at $t = n$ for every n divisible by 4.

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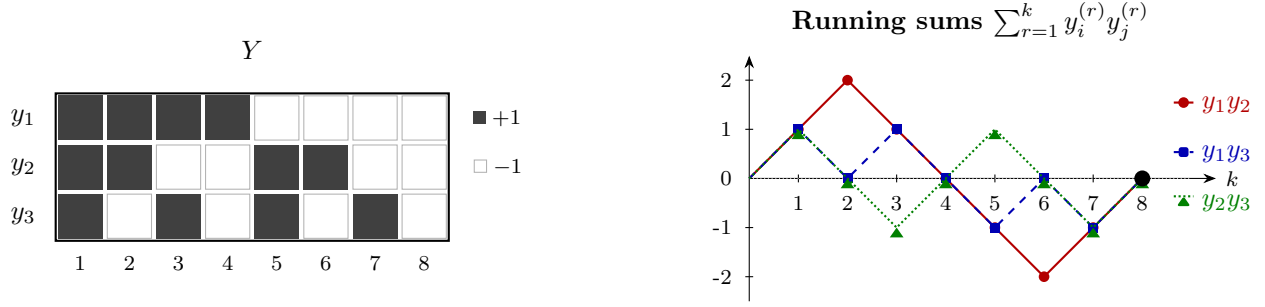


Figure 1: A 3×8 partial Hadamard matrix (left) and the running pairwise-product sums (right). The matrix is partial Hadamard exactly when all three sums return to zero at $k = 8$.

Using this framework, De Launey and Levin [?, Theorem 2] proved that for every fixed $n \geq 2$, the number of $n \times 4t$ partial Hadamard matrices satisfies

$$N_{n,4t} = [1 + o(1)] A_{n,4t} \quad \text{as } t \rightarrow \infty, \quad (3)$$

where $A_{n,s} := 2^{ns+2d-n+1}(2\pi s)^{-d/2}$. This gives both existence for large t and a precise asymptotic count. Since the Hadamard conjecture is equivalent to $N_{n,n} > 0$ whenever n is divisible by 4, one can also ask what happens when n grows with t , potentially at a slower rate: does $N_{n,4t} \sim A_{n,4t}$ still hold, and if so, with what corrections? Two prior works address this in the regime $t/n^\alpha \rightarrow \infty$:

- (i) De Launey and Levin's proofs [?, Theorem 4.1] give $N_{n,4t} \sim A_{n,4t}$ as $t/n^{12} \rightarrow \infty$, though this is never explicitly stated.
- (ii) Canfield [?] established the same asymptotic as $t/n^4 \rightarrow \infty$ in unpublished work.

Our main result is a first-order expansion that extends these to $\alpha = 3$ and reveals the leading correction:

Theorem 1.1. *There exist $C_0, c > 0$ such that for all sufficiently large n and $t \geq C_0 n^3$,*

$$\frac{N_{n,4t}}{A_{n,4t}} = 1 - \frac{\binom{n}{3}}{8t} + O\left(\frac{n^2}{t} + \frac{n^{5/2}}{t^{3/2}} + \frac{n^6}{t^2} + e^{-cn^2}\right).$$

The interesting regime is $n \rightarrow \infty$ with $t \geq C_0 n^3$, since fixed n is already covered by [?]. When $t/n^3 \rightarrow \infty$ all corrections vanish, recovering $N_{n,4t} \sim A_{n,4t}$ (Corollary 4.1). When $t = \Theta n^3$ with $\Theta \geq C_0$ fixed, the leading correction is $\binom{n}{3}/(8\Theta n^3) \approx 1/(48\Theta)$; the remaining error is $O(\Theta^{-2} + 1/n + e^{-cn^2})$, so for Θ large and n large the correction dominates the error. Thus the expansion captures the first deviation of $N_{n,4t}$ from the scale $A_{n,4t}$. Below $\alpha = 3$ the asymptotics of $N_{n,4t}$ are open (Figure 2).

As in [?], the proof proceeds by estimation of the Fourier integral (2), split into a *primary* contribution from neighborhoods of the points where $|\psi| = 1$ and a *residual* over the rest of the torus. Beyond improving the exponent, a motivation for pushing the Fourier-analytic approach is that it provides a direct, self-contained route to the counting asymptotics, and it is valuable to understand the limits of these tools. We describe the proof structure below, then compare with the approaches of [?] and [?] in Section 1.2.

1.1 Proof overview

Figure 3 illustrates the decomposition. Since partial Hadamard matrices with $n \geq 3$ rows exist only when the number of columns is divisible by 4, we write the number of columns as $4t$ throughout;

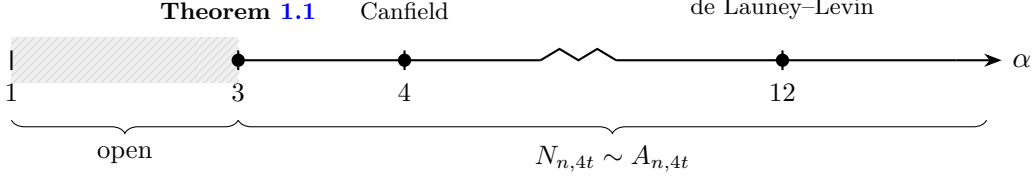


Figure 2: Each marker indicates the smallest α for which the corresponding work establishes $N_{n,4t} \sim A_{n,4t}$ as $t/n^\alpha \rightarrow \infty$. Corollary 4.1 extends this to $\alpha = 3$. Theorem 1.1 further shows that $N_{n,4t}/A_{n,4t}$ has a nonvanishing correction when t/n^3 converges to a constant.

the integral to evaluate is

$$\mathbb{P}(S_{4t} = 0) = (2\pi)^{-d} \int_{\mathbb{T}^d} \psi(\lambda)^{4t} d\lambda. \quad (4)$$

Integral decomposition. The integrand $\psi(\lambda)^{4t}$ is largest near the lattice

$$\Lambda = \{\lambda \in \mathbb{T}^d : |\psi(\lambda)| = 1\}.$$

As shown in [?, Lemmas 2.2–2.3], each coordinate of every $\lambda \in \Lambda$ lies in $\{0, \pm\pi/2, \pi\}$, $|\Lambda| = 2^{2d-n+1}$, and $\psi(\lambda)^{4t} = 1$ for all $\lambda \in \Lambda$. We place a small box $\mathcal{B}_\delta(\lambda) = \lambda + [-\delta, \delta]^d$ around each lattice point (Figure 3(a), blue) and call the remainder $R_\delta = \mathbb{T}^d \setminus \bigcup_{\lambda \in \Lambda} \mathcal{B}_\delta(\lambda)$ the *residual*.

A multiplicative identity, $\psi(\lambda + \mu) = \psi(\lambda)\psi(\mu)$ for $\lambda \in \Lambda$ and $\mu \in [-\pi/4, \pi/4]^d$, together with $\psi(\lambda)^{4t} = 1$, shows that the integral over every primary box equals the integral over the box at the origin. Summing over Λ gives the primary-secondary decomposition [?, Proposition 2.1]:

$$\mathbb{P}(S_{4t} = 0) = \underbrace{|\Lambda| (2\pi)^{-d} \int_{\mathcal{B}_\delta} \psi(\mu)^{4t} d\mu}_{\text{primary}} + \underbrace{(2\pi)^{-d} \int_{R_\delta} \psi(\gamma)^{4t} d\gamma}_{\text{residual}}. \quad (5)$$

Writing $K_n := |\Lambda|(2\pi)^{-d} = 2^{2d-n+1}(2\pi)^{-d}$ and splitting the primary integral at the core boundary $\mathcal{D}_{\text{core}} = \{\|\mu\|^2 \leq d/t\}$ refines this into a three-term decomposition:

$$\mathbb{P}(S_{4t} = 0) = \underbrace{K_n \int_{\mathcal{D}_{\text{core}}} \psi(\mu)^{4t} d\mu}_{\text{core}} + \underbrace{K_n \int_{\mathcal{B}_\delta \setminus \mathcal{D}_{\text{core}}} \psi(\mu)^{4t} d\mu}_{\text{off-core}} + \underbrace{(2\pi)^{-d} \int_{R_\delta} \psi(\gamma)^{4t} d\gamma}_{\text{residual}}. \quad (6)$$

Since $\mathbb{P}(S_{4t} = 0)$ is real, the proof reduces to three estimates, where $\hat{A}_{n,4t} := A_{n,4t} \cdot 2^{-4nt}$:

- (i) the core contributes $(1 - \binom{n}{3}/(8t) + \text{lower order}) \hat{A}_{n,4t}$ for $t \geq C_0 n^3$ (Section 2),
- (ii) the off-core contributes $o(\hat{A}_{n,4t})$ as $t/(n^{8/3} \log t) \rightarrow \infty$ (Section 3),
- (iii) the residual contributes $o(\hat{A}_{n,4t})$ as $t/(n^{8/3} \log t) \rightarrow \infty$ (Section 3).

Primary term. Near the origin, ψ is close to a Gaussian characteristic function. More precisely, expanding $\log \psi(\mu)$ to sixth order produces a factorization

$$\psi(\mu)^{4t} = \underbrace{e^{-2t\|\mu\|^2}}_{\text{Gaussian}} \cdot \underbrace{e^{-4itT(\mu)}}_{\text{cubic phase}} \cdot (\text{quartic, quintic, and sixth-order terms}),$$

where $T(\mu) = \sum_{\text{triangles } \tau} \prod_{e \in \tau} \mu_e$ is a cubic form summing over the $\binom{n}{3}$ triangles of the complete graph on n vertices. The Gaussian factor $e^{-2t\|\mu\|^2}$ concentrates the integrand on the *core* $\mathcal{D}_{\text{core}} =$

$\{\|\mu\|^2 \leq d/t\}$ (Figure 3(c), red), where the higher-order terms are small. Outside the core (on the *annulus* and the *corners*, Figure 3(c), yellow and blue), the Gaussian decay makes the contribution exponentially negligible.

The main difficulty on the core is the cubic phase $\exp(-4itT(\mu))$, which oscillates and does not vanish pointwise. We integrate it directly: because T is an odd function ($T(-\mu) = -T(\mu)$) and $\mathcal{D}_{\text{core}}$ is centrally symmetric, the imaginary part of $\int_{\mathcal{D}_{\text{core}}} e^{-2t\|\mu\|^2} e^{-4itT(\mu)} d\mu$ vanishes. The loss in the real part is controlled by $1 - \cos(4tT) \leq 8t^2T^2$; Gaussian moment estimates on T^2 then give

$$\operatorname{Re} \int_{\mathcal{D}_{\text{core}}} e^{-2t\|\mu\|^2} e^{-4itT(\mu)} d\mu \geq [1 - O(n^3/t)] G_{\text{core}}(d, t),$$

where $G_{\text{core}}(d, t) = \int_{\mathcal{D}_{\text{core}}} e^{-2t\|\mu\|^2} d\mu$. The quartic and quintic terms are controlled in L^2 against the Gaussian weight using hypercontractivity [?, Theorem 9.22] and a vertex-peeling argument; the sixth-order remainder is bounded by Gaussian moment estimates. Altogether, the primary integral equals $[1 - \binom{n}{3}/(8t) + O(n^2/t + n^{5/2}/t^{3/2} + n^6/t^2 + e^{-cn^2})] \hat{A}_{n,4t}$ for $t \geq C_0 n^3$. The correction $\binom{n}{3}/(8t)$ vanishes as $t/n^3 \rightarrow \infty$ but is bounded away from zero when $t \asymp n^3$.

Off-core and residual. It remains to show that the off-core integral over $\mathcal{B}_\delta \setminus \mathcal{D}_{\text{core}}$ and the residual integral over R_δ are both negligible compared to $\hat{A}_{n,4t}$. These regions decompose into pieces with decreasing ease of control (Figure 3(a)–(b)). The key pointwise tool is the following bound from [?].

Lemma 1.2 (Cosine-product bound [?, Lemma 3.1]). *For every $\gamma \in [-\pi, \pi]^d$ and every $k \in \{1, \dots, n\}$, $|\psi(\gamma)|^2 \leq \frac{1}{2} + \frac{1}{2} \prod_{i \neq k} \cos(2\gamma_{\{i,k\}})$.*

Odd cells. The superlattice $\Lambda_0 = \{\lambda \in \mathbb{T}^d : \text{every coordinate} \in \{0, \pm\pi/2, \pi\}\}$ tiles the torus into quarter-boxes $\mathcal{B}_{\pi/4}(\lambda)$. All but a 2^{1-n} -fraction of these cells correspond to lattice points in $\Lambda_0 \setminus \Lambda$, where the cosine-product bound (Lemma 1.2) gives $|\psi(\gamma)|^2 \leq 1/2$. On these cells $|\psi|^{4t} \leq 2^{-2t}$, so their total contribution is $o(\hat{A}_{n,4t})$.

Near shell. Each even cell $\mathcal{B}_{\pi/4}(\lambda)$ contains the primary box $\mathcal{B}_\delta(\lambda)$. Let $r \in (0, \pi/4)$ be a small absolute constant (fixed in Section 3) such that the cumulant expansion of $\log \psi$ is valid on the ball $\mathcal{D}_r = \{\lambda : \|\lambda\|^2 \leq r^2\}$. On the region $\mathcal{D}_r \setminus \mathcal{B}_\delta$ between the primary box and this ball (Figure 3(b), orange), the norm $\|\mu\|$ exceeds δ , and the small-ball decay of ψ from the cumulant expansion gives $|\psi(\mu)|^{4t} \leq \exp(-ct\delta^2)$. With $\delta^2 = 2d/t$, this contributes at most $\exp(-cd) \hat{A}_{n,4t}$.

Far shell. On the remaining region $\mathcal{B}_{\pi/4} \setminus \mathcal{D}_r$ (Figure 3(b)–(c)), we need a pointwise bound $|\psi(\lambda)| < 1$ strong enough to overwhelm the volume $(\pi/2)^d$. The cosine-product bound (Lemma 1.2) provides such a contraction when the *row-sums* $I_k(\lambda) := \sum_{i \neq k} \lambda_{\{i,k\}}^2$ are large for some k , but it is too weak when all row-sums are small.

To handle this sub-region, we replace the Rademacher entries ξ_i with i.i.d. standard Gaussians g_i and write $\psi_G(\lambda) = \mathbb{E}[\exp(i \sum_{i < j} \lambda_{\{i,j\}} g_i g_j)]$ for the resulting characteristic function. The key property is $|\psi_G(\lambda)| \leq (1 + 2\|\lambda\|^2)^{-1/4}$, which contracts as soon as $\|\lambda\|$ is bounded away from zero, regardless of how the norm is distributed across coordinates. A Gaussian comparison inequality (Corollary 3.2) bounds the distance:

$$|\psi(\lambda) - \psi_G(\lambda)| \leq \frac{3}{2} \sum_{k=1}^n I_k(\lambda)^{3/2}.$$

When all row-sums are small, this error is small and $|\psi|$ inherits the contraction of $|\psi_G|$. When some row-sum I_k is large, the cosine-product bound $|\psi(\lambda)|^2 \leq \frac{1}{2} + \frac{1}{2} \prod_{i \neq k} \cos(2\lambda_{\{i,k\}})$ already

gives a strict contraction through the exponential decay $\cos(2x) \leq \exp(-8x^2/\pi^2)$. In both cases, $|\psi| \leq q < 1$ for a constant q independent of n , so the contribution decays exponentially in t .

Between these two regimes lies an intermediate region where the comparison error is too large for the Gaussian bound but the row-sums are too small for the cosine-product bound to give an n -independent rate. Here the contraction $|\psi|^{4t} \leq \exp(-ctn^{-2/3})$ is n -dependent, but it still overwhelms the volume $(\pi/2)^d$ because $tn^{-2/3} \gg d \log t \asymp n^2 \log t$ whenever $t \gg n^{8/3} \log t$. This is the tightest constraint from the residual, but it is well below the $t \geq C_0 n^3$ constraint required by the primary analysis. The primary term, not the residual, determines the exponent $\alpha = 3$.

1.2 Comparison with prior work

All three proofs ([?], [?], and the present work) use the primary/residual decomposition (5). The exponent α is determined by how two difficulties are resolved: the cubic phase on the primary boxes, and the pointwise decay on the far shell. We describe each approach in the notation introduced above.

The cubic phase and the primary integral. On the primary box $\mathcal{B}_\delta$, the characteristic function ψ is close to Gaussian but has a non-trivial cubic phase driven by the triangle form T . The three proofs handle this phase differently.

De Launey and Levin [?] separate ψ into real and imaginary parts. On $\mathcal{B}_\delta$ the ratio $\beta := \text{Im}(\psi)/\text{Re}(\psi)$ is at most $C(n\delta)^3$, and they bound $\text{Re}(\psi^{4t})$ below by $\text{Re}(\psi)^{4t}$. This lower bound requires $4t\beta < 1$: if β is too large relative to $1/t$, raising ψ to the $4t$ -th power amplifies the imaginary part and destroys the bound. Since $\beta \leq C(n\delta)^3$, the constraint forces $\delta \leq C(tn^3)^{-1/3}$, shrinking the primary box as t and n grow. However, shrinking δ enlarges the residual region, which is harder to control (see below).

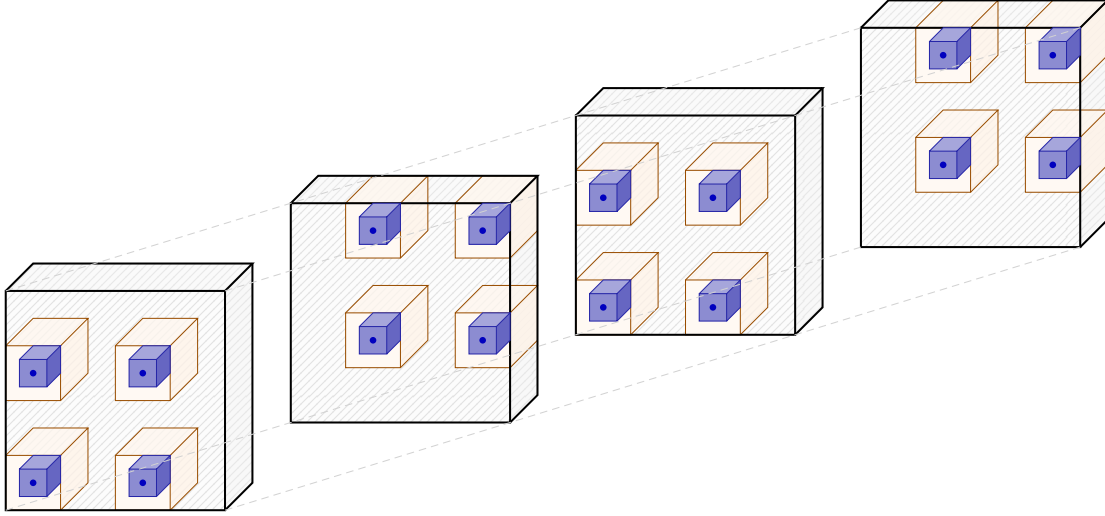
Canfield [?] avoids this hard constraint by integrating one coordinate at a time. The cubic phase can be written $T(\mu) = \sum_{j < k} \mu_{jk} B_{jk}$, where $B_{jk} := \sum_\ell \mu_{j\ell} \mu_{k\ell}$ sums over the remaining edges of triangles through $\{j, k\}$. Because μ_{jk} couples linearly to B_{jk} , completing the square $\exp(-t\mu_{jk}^2/2 - it\mu_{jk} B_{jk}) = \exp(-tB_{jk}^2/2) \cdot \exp(-t(\mu_{jk} + iB_{jk})^2/2)$ reduces each one-dimensional integral to a Gaussian at the cost of a factor $\exp(-tB_{jk}^2/2)$. After all d steps, the accumulated error is $d \cdot O(n^2/t) = O(n^4/t)$, giving $\alpha = 4$. Canfield identifies $\alpha = 3$ as the natural target but leaves the required cubic integration as an open problem (see [?]).

Our approach integrates the cubic phase in one step rather than absorbing it variable by variable. As described in Section 1.1, confining the analysis to the core $\mathcal{D}_{\text{core}}$ and exploiting the antisymmetry of T gives an error of $O(n^3/t)$, one power of n better than Canfield's accumulated error. Because the primary lower bound lives entirely on $\mathcal{D}_{\text{core}}$, the remainder of $\mathcal{B}_\delta$ is bounded as part of the off-core estimates (6), and the choice of δ is decoupled from the primary analysis.

The residual and the far shell. De Launey and Levin bound the entire residual R_δ by the uniform estimate $|\psi(\gamma)| \leq \cos \delta \leq \exp(-(11/24)\delta^2)$, yielding $|(2\pi)^{-d} \int_{R_\delta} \psi^{4t}| \leq \exp(-(11/6)t\delta^2)$. For this to be $o(\hat{A}_{n,4t})$ one needs $t\delta^2 \gg d \log t$. Combined with the constraint $\delta \leq C(tn^3)^{-1/3}$ from their primary analysis, this gives $t^{1/3}/n^2 \gg n^2 \log t$, hence $t \gg n^{12}$. Canfield improves the residual bound by a factor of n in the exponent, but this is not his bottleneck.

In our proof, the odd cells and the near shell $\mathcal{D}_r \setminus \mathcal{B}_\delta$ are negligible by elementary bounds (Section 1.1). The critical piece is the far shell $\mathcal{B}_{\pi/4} \setminus \mathcal{D}_r$. When all row-sums I_k are small, the cosine-product bound (Lemma 1.2) decays at an n -dependent rate that is too weak; the Gaussian

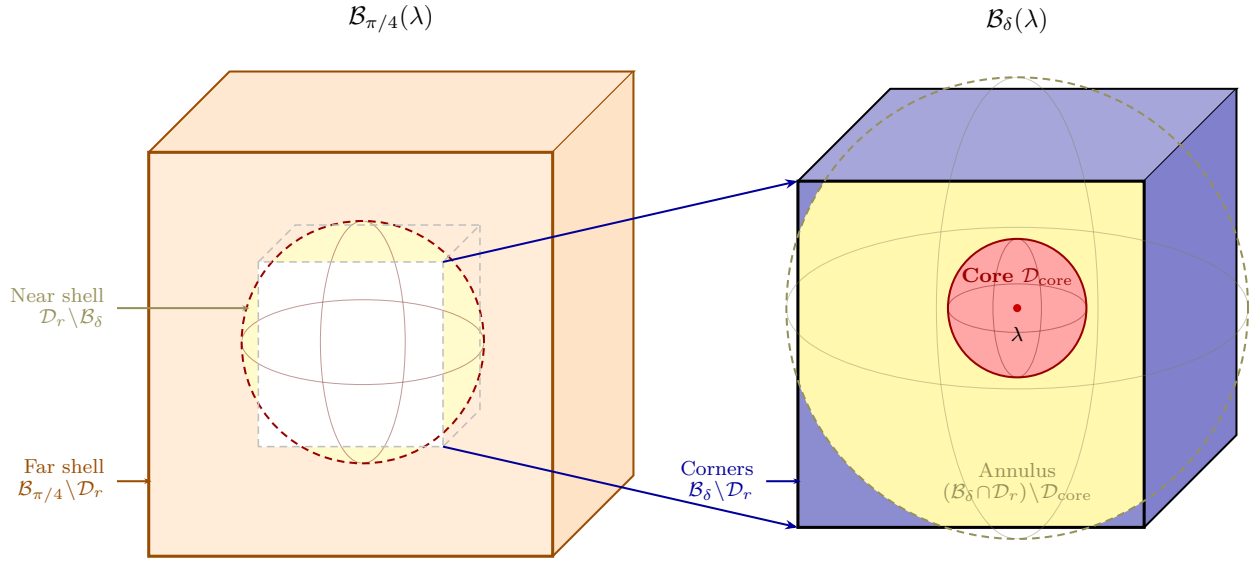
$$\mathbb{T}^3 \quad (n = 3, d = 3)$$



$$\bullet \Lambda \quad \blacksquare \mathcal{B}_\delta(\lambda) \quad \blacksquare \mathcal{B}_{\pi/4}(\lambda) \quad \text{hatched } R_\delta^{\text{odd}}$$

$$R_\delta = \mathbb{T}^d \setminus \bigcup_{\lambda \in \Lambda} \mathcal{B}_\delta(\lambda) = \text{orange } R_\delta^{\text{even}} \cup R_\delta^{\text{odd}}$$

(a) Torus decomposition



(b) Even cell

(c) Primary box

Figure 3: Composite overview. (a) The torus $\mathbb{T}^3$ decomposes into primary boxes $\mathcal{B}_\delta$ (blue), even residual cells (orange), and odd cells (hatched). (b) A single even cell $\mathcal{B}_{\pi/4}$; the blue $\mathcal{B}_\delta$ is expanded in (c). (c) Inside $\mathcal{B}_\delta$: the core $\mathcal{D}_{\text{core}}$ (red), annulus (yellow), and corners (blue).

comparison inequality (Corollary 3.2) provides the n -independent contraction needed in this sub-region.

The progression from $\alpha = 12$ to $\alpha = 3$ is driven by a decoupling of the primary and residual estimates. De Launey and Levin's two-way split forces $\delta \leq C(tn^3)^{-1/3}$ on the primary side while the residual needs $t\delta^2$ large. Canfield's coordinate-by-coordinate integration frees the residual from δ but accumulates $O(n^4/t)$ error. The three-way decomposition removes this coupling: the Gaussian comparison resolves the far-shell bottleneck, and all thresholds align at $t \asymp n^3$. Pushing below $\alpha = 3$ would require finer control of the cubic phase on the core, since the residual is already negligible for $t \gg n^{8/3} \log t$.

Acknowledgement of AI tools

The author used AI tools throughout this project for literature search, editing, and figure generation. More substantially, the author built a custom harness around GPT 5.4 Pro that identified bottlenecks in the existing proof approaches and, after considerable iteration, helped guide the analysis to $\alpha = 3$. Details of this process and harness will be discussed elsewhere.

1.3 Notation and preliminaries

We fix notation and collect the preliminary results used in Sections 2–4. We include short proofs of some of the results from [?] for completeness.

Notation. We retain n , $d = \binom{n}{2}$, $Z(y)$, ξ , ψ , $N_{n,s}$, Λ , $\mathcal{B}_\delta$, R_δ , $\hat{A}_{n,4t}$, and $A_{n,4t}$ as defined in Section 1.1. For i.i.d. copies $\xi^{(1)}, \dots, \xi^{(s)}$ of ξ , write $S_s = \sum_{r=1}^s Z(\xi^{(r)})$.

For the cumulant expansion we write $X_\lambda = \lambda \cdot Z(\xi)$, $s(\lambda) = \|\lambda\|^2$, $\kappa_r(\lambda)$ for the r -th cumulant of X_λ , $T(\lambda) = \kappa_3(\lambda)/6$, $Q(\lambda) = \kappa_4(\lambda)/24$, and $P(\lambda) = \kappa_5(\lambda)/120$. For $r > 0$, set $\mathcal{D}_r = \{\lambda \in \mathbb{R}^d : \|\lambda\|^2 \leq r^2\}$. The core region, the core Gaussian mass, and the full Gaussian integral are

$$\mathcal{D}_{\text{core}} = \left\{ \lambda \in \mathbb{R}^d : \|\lambda\|^2 \leq \frac{d}{t} \right\}, \quad G_{\text{core}}(d, t) = \int_{\mathcal{D}_{\text{core}}} e^{-2t\|\lambda\|^2} d\lambda, \quad F(d, t) = \left(\frac{\pi}{2t} \right)^{d/2}.$$

Results from [?]. The lattice Λ has a rigid algebraic structure, which we use to decompose the integral.

Lemma 1.3 (Lattice structure). *The following hold.*

- (i) If $\lambda \in \Lambda$ and $\gamma \in \mathcal{B}_{\pi/4}$, then $\psi(\lambda + \gamma) = \psi(\lambda)\psi(\gamma)$.
- (ii) Every coordinate of every $\lambda \in \Lambda$ lies in $\{0, \pm\pi/2, \pi\}$.
- (iii) If $\delta < \pi/4$, then the boxes $\mathcal{B}_\delta(\lambda)$, $\lambda \in \Lambda$, are pairwise disjoint.
- (iv) The multiset $\{\psi(\lambda) : \lambda \in \Lambda\}$ consists of $\pm 1, \pm i$, each with multiplicity 2^{2d-n-1} . Consequently $\psi(\lambda)^{4t} = 1$ for $\lambda \in \Lambda$, and $|\Lambda| = 2^{2d-n+1}$.
- (v) Each $\lambda \in \Lambda_0$ decomposes uniquely as $\lambda^{(1)} + \lambda^{(2)}$ with $\lambda^{(1)} \in \{0, \pi\}^d$ and $\lambda^{(2)} \in \{0, \pi/2\}^d$. Associate to $\lambda^{(2)}$ the graph $G_{\lambda^{(2)}}$ on $[n]$ with edge $\{i, j\}$ whenever $\lambda_{\{i, j\}}^{(2)} = \pi/2$. Then $\lambda \in \Lambda$ if and only if every vertex of $G_{\lambda^{(2)}}$ has even degree.

The first two parts are [?, Lemma 2.2], part (iii) follows immediately from part (ii), part (iv) is [?, Lemma 2.3], and part (v) is [?, Lemma 2.4].

The multiplicative identity (i) and the fact that every $\psi(\lambda)^4 = 1$ for $\lambda \in \Lambda$ (iv) justify the decomposition (5) from the proof overview.

Proposition 1.4 (Primary-secondary decomposition [?, Proposition 2.1]). *For every $t \geq 1$ and every $0 < \delta < \pi/4$,*

$$\mathbb{P}(S_{4t} = 0) = 2^{2d-n+1}(2\pi)^{-d} \int_{\mathcal{B}_\delta} \psi(\lambda)^{4t} d\lambda + (2\pi)^{-d} \int_{R_\delta} \psi(\gamma)^{4t} d\gamma.$$

The residual R_δ further splits into two types of cells, distinguished by the parity of the underlying lattice point. The odd cells are negligible by an elementary sign argument; the even cells require more delicate bounds.

Lemma 1.5 (Residual decomposition [?, Lemma 2.7]). *For $0 < \delta < \pi/4$,*

$$R_\delta = R_\delta^{\text{even}} \cup R_\delta^{\text{odd}}, \quad R_\delta^{\text{even}} = \bigcup_{\lambda \in \Lambda} (\lambda + [\mathcal{B}_{\pi/4} \setminus \mathcal{B}_\delta]), \quad R_\delta^{\text{odd}} = R_\delta \setminus R_\delta^{\text{even}},$$

and the sets in the union are disjoint.

Proof. Since every coordinate of every $\lambda \in \Lambda$ lies in $\{0, \pm\pi/2, \pi\}$ (Lemma 1.3(ii)), the translates $\mathcal{B}_{\pi/4}(\lambda)$, $\lambda \in \Lambda_0$, tile $\mathbb{T}^d$, where Λ_0 is the set of all points with coordinates in $\{0, \pm\pi/2, \pi\}$. The lattice Λ is a subset of Λ_0 . Each tile centered at a point in Λ contributes $\mathcal{B}_{\pi/4}(\lambda) \setminus \mathcal{B}_\delta(\lambda)$ to R_δ^{even} , and each tile centered at a point in $\Lambda_0 \setminus \Lambda$ contributes a full quarter-box to R_δ^{odd} . $\square$

On odd cells the characteristic function is bounded by a constant less than one, making the contribution of these cells exponentially small. The following bound is implicit in the proof of [?, Proposition 3.2]; we give a self-contained argument.

Lemma 1.6 (Odd-cell bound). *If $\gamma \in R_\delta^{\text{odd}}$, then $|\psi(\gamma)|^2 \leq \frac{1}{2}$.*

Proof. It suffices to show that some $k \in \{1, \dots, n\}$ satisfies $\prod_{i \neq k} \cos(2\gamma_{\{i,k\}}) \leq 0$, since the cosine-product bound (Lemma 1.2) then gives $|\psi(\gamma)|^2 \leq \frac{1}{2}$.

By Lemma 1.5, γ lies in a quarter-box $\mathcal{B}_{\pi/4}(\lambda)$ with $\lambda \in \Lambda_0 \setminus \Lambda$. Write $\lambda = \lambda^{(1)} + \lambda^{(2)}$, where each coordinate of $\lambda^{(1)}$ lies in $\{0, \pi\}$ and each coordinate of $\lambda^{(2)}$ lies in $\{0, \pi/2\}$. By Lemma 1.3(v), $\lambda \in \Lambda$ if and only if every vertex of $G_{\lambda^{(2)}}$ has even degree. Since $\lambda \notin \Lambda$, some vertex k has odd degree, i.e., an odd number of the coordinates $\lambda_{\{i,k\}}^{(2)}$, $i \neq k$, equal $\pi/2$.

Write $\gamma = \lambda^{(1)} + \lambda^{(2)} + \mu$ with $\mu \in \mathcal{B}_{\pi/4}$. For each $i \neq k$,

$$\cos(2\gamma_{\{i,k\}}) = \cos(2\lambda_{\{i,k\}}^{(1)} + 2\lambda_{\{i,k\}}^{(2)} + 2\mu_{\{i,k\}}).$$

Since $2\lambda_{\{i,k\}}^{(1)} \in \{0, 2\pi\}$ and $2\lambda_{\{i,k\}}^{(2)} \in \{0, \pi\}$, this equals $(-1)^{b_{ik}} \cos(2\mu_{\{i,k\}})$ with $b_{ik} \in \{0, 1\}$, and an odd number of the b_{ik} equal 1. Because $|\mu_{\{i,k\}}| \leq \pi/4$, each factor $\cos(2\mu_{\{i,k\}}) \geq 0$, so the product $\prod_{i \neq k} \cos(2\gamma_{\{i,k\}})$ carries an odd number of sign flips and is therefore nonpositive. $\square$

The sixth-order expansion of $\log \psi$ (Lemma 2.6) requires that $\text{Re } \psi$ stay bounded away from zero along a path, so that the principal branch of the logarithm is well-defined. The following bound, which is immediate from [?, Lemma 3.1, eq. (3.2)], provides this.

Lemma 1.7 (Positivity of $\text{Re } \psi$ [?, Lemma 3.1]). *If $s(\lambda) \leq 1/2$, then $\text{Re } \psi(\lambda) \in [3/4, 1]$.*

Proof. Since $\text{Re } \psi(\lambda) = \mathbb{E}[\cos X_\lambda]$, the bound $\cos x \geq 1 - x^2/2$ gives $\text{Re } \psi(\lambda) \geq 1 - \frac{1}{2}s(\lambda) \geq 3/4$. $\square$

Our core and off-core analysis (Sections 2–3) yields asymptotics as $n \rightarrow \infty$. For bounded n , the following fixed- n result from [?] closes the argument in Section 4.

Lemma 1.8 (Fixed- n asymptotics [?, Theorem 4.1]). *For every fixed integer $n \geq 2$, $N_{n,4t} = [1 + o_{t \rightarrow \infty}(1)] A_{n,4t}$ as $t \rightarrow \infty$.*

Standard analytic tools. The cumulant expansion on the core produces polynomial corrections to the Gaussian integrand. We control these using three standard estimates.

Lemma 1.9 (Hypercontractive inequality [?, Theorems 9.22 and 11.23]). *Let H be a polynomial of degree at most q in either independent Rademacher signs or independent centered Gaussians. Then for every $p \geq 2$, $\|H\|_{L^p} \leq (p-1)^{q/2} \|H\|_{L^2}$.*

Hypercontractivity converts L^p norms of Q and P into L^2 norms, which are explicit. The following moment bound controls the sixth-order remainder against the Gaussian weight.

Lemma 1.10 (Gaussian radial moments). *For every integer $m \geq 0$ there is a constant $C_m > 0$ such that $\int_{\mathbb{R}^d} \|\lambda\|^{2m} e^{-2t\|\lambda\|^2} d\lambda \leq C_m (d/t)^m F(d, t)$ for all $d, t \geq 1$.*

The following identity is used in two ways: the real case drives is used in vertex-peeling argument in the quartic bounds (Proposition 2.4), and the complex case gives a determinant formula for the Gaussian characteristic function ψ_G in the far-shell decomposition (Proposition 3.3).

Lemma 1.11 (Gaussian quadratic integral). *Let A be a complex symmetric $m \times m$ matrix whose real part is positive definite. Then $\int_{\mathbb{R}^m} e^{-x^\top A x} dx = \pi^{m/2} \det(A)^{-1/2}$. In particular, if $g \sim N(0, I_m)$ and M is real symmetric, taking $A = \frac{1}{2}(I - iM)$ gives $\mathbb{E}[e^{ig^\top M g/2}] = \det(I - iM)^{-1/2}$ and $|\mathbb{E}[e^{ig^\top M g/2}]| = \det(I + M^2)^{-1/4}$.*

Proof. Write $A = \text{Re}(A) + i \text{Im}(A)$. The substitution $y = \text{Re}(A)^{1/2} x$ gives

$$\int_{\mathbb{R}^m} e^{-x^\top A x} dx = \det(\text{Re } A)^{-1/2} \int_{\mathbb{R}^m} e^{-\|y\|^2 - iy^\top S y} dy,$$

where $S = \text{Re}(A)^{-1/2} \text{Im}(A) \text{Re}(A)^{-1/2}$ is real symmetric. Diagonalizing $S = O^\top \text{diag}(s_1, \dots, s_m) O$ with O orthogonal and setting $z = O y$, the integral factors as

$$\prod_{j=1}^m \int_{\mathbb{R}} e^{-(1+is_j)z_j^2} dz_j = \prod_{j=1}^m \sqrt{\frac{\pi}{1+is_j}} = \pi^{m/2} \det(I + iS)^{-1/2}.$$

Since $A = \text{Re}(A)^{1/2}(I + iS) \text{Re}(A)^{1/2}$, we have $\det(A) = \det(\text{Re } A) \det(I + iS)$, and the result follows. $\square$

Auxiliary lemma. The following estimate shows that the Gaussian integral over the core $\mathcal{D}_{\text{core}}$ captures nearly all of the full Gaussian integral $F(d, t) = \int_{\mathbb{R}^d} e^{-2t\|\lambda\|^2} d\lambda = (\pi/(2t))^{d/2}$.

Lemma 1.12 (Core Gaussian mass). *There exists an absolute constant $c_* > 0$ such that $G_{\text{core}}(d, t) \geq (1 - e^{-c_* d}) F(d, t)$.*

Proof. On the region $\|\lambda\|^2 > d/t$, one has $t\|\lambda\|^2 > d$, hence $e^{-2t\|\lambda\|^2} \leq e^{-d} e^{-t\|\lambda\|^2}$. Therefore

$$G_{\text{core}}(d, t) = F(d, t) - \int_{\|\lambda\|^2 > d/t} e^{-2t\|\lambda\|^2} d\lambda \geq F(d, t) - e^{-d} \int_{\mathbb{R}^d} e^{-t\|\lambda\|^2} d\lambda = (1 - e^{-c_* d}) F(d, t),$$

where $c_* = 1 - \frac{1}{2} \log 2 > 0$. $\square$

2 Core

This section establishes the primary estimate: Proposition 2.9 gives a first-order expansion of $K_n \operatorname{Re} \int_{\mathcal{D}_{\text{core}}} \psi^{4t}$ with leading correction $-(\binom{n}{3})/(8t)$ for $t \geq C_0 n^3$ and n large. Combined with the off-core upper bounds in Section 3, this proves the main counting theorem.

The argument expands $\log \psi$ to fifth order on $\mathcal{D}_{\text{core}}$, isolates the cubic phase $T(\lambda)$, and shows that Q , P , and E_6 are negligible when $t \gg n^3$. The imaginary part of the cubic phase vanishes by antisymmetry of T , and the real part differs from the Gaussian integral $G_{\text{core}}(d, t)$ by $O(n^3/t)$.

2.1 Cumulants for the log expansion

We compute the low-order cumulants of X_λ entering the Taylor expansion of $\log \psi(\lambda) = \log \mathbb{E}[e^{iX_\lambda}]$. The identities below produce the explicit cubic, quartic, and quintic phase terms T , Q , and P .

The first identity is implicit in the proof of [?, Lemma 3.1]; we state it separately for later reference.

Lemma 2.1 (Triangle formula). *For every $\lambda \in \mathbb{R}^d$, $T(\lambda) = \sum_{1 \leq i < j < k \leq n} \lambda_{ij} \lambda_{ik} \lambda_{jk}$.*

Proof. Expanding the cube gives

$$\mathbb{E}[X_\lambda^3] = \sum_{e_1, e_2, e_3} \lambda_{e_1} \lambda_{e_2} \lambda_{e_3} \mathbb{E}[\chi_{e_1} \chi_{e_2} \chi_{e_3}],$$

where $\chi_{\{i,j\}} = \xi_i \xi_j$. The expectation vanishes unless every vertex index appears with even multiplicity. With three edges and no loops, the only such multigraph is a triangle on three distinct vertices. Thus the only surviving ordered triples are the $3!$ orderings of $(\{i, j\}, \{i, k\}, \{j, k\})$ for $1 \leq i < j < k \leq n$. For each such ordering, $(\xi_i \xi_j)(\xi_i \xi_k)(\xi_j \xi_k) = 1$. Summing over all triangles and their $3!$ orderings gives

$$\mathbb{E}[X_\lambda^3] = 6 \sum_{1 \leq i < j < k \leq n} \lambda_{ij} \lambda_{ik} \lambda_{jk}.$$

Since $T(\lambda) = \frac{1}{6} \mathbb{E}[X_\lambda^3]$ by definition, the claim follows. $\square$

Define

$$Q(\lambda) := \frac{\kappa_4(\lambda)}{24}.$$

The next lemma gives an explicit formula for Q . Write

$$\mathcal{C}_4(\lambda) = \sum_{\substack{i_1, i_2, i_3, i_4 \\ \text{distinct}}} \lambda_{\{i_1, i_2\}} \lambda_{\{i_2, i_3\}} \lambda_{\{i_3, i_4\}} \lambda_{\{i_4, i_1\}} \quad (7)$$

for the sum over ordered 4-cycles.

Lemma 2.2 (Fourth cumulant).

$$Q(\lambda) = -\frac{1}{12} \sum_e \lambda_e^4 + \frac{1}{8} \mathcal{C}_4(\lambda). \quad (8)$$

Proof. Since $\kappa_4(\lambda) = \mathbb{E}[X_\lambda^4] - 3s(\lambda)^2$, we expand $\mathbb{E}[X_\lambda^4]$ by writing four copies of $X_\lambda = \sum_e \lambda_e \chi_e$ and keeping only terms where every vertex has even multiplicity:

- (i) All four edges equal: contributes $\sum_e \lambda_e^4$.
- (ii) Two distinct edges, each chosen twice: $\binom{4}{2} = 6$ orderings, contributing $6 \sum_{e < f} \lambda_e^2 \lambda_f^2$.

(iii) Four distinct edges forming an undirected 4-cycle: each cycle monomial $\lambda_{ab}\lambda_{bc}\lambda_{cd}\lambda_{da}$ appears $4! = 24$ times in the raw expansion and 8 times in $\mathcal{C}_4$ (two cyclic orientations, four starting points), giving coefficient $24/8 = 3$.

Therefore $\mathbb{E}[X_\lambda^4] = \sum_e \lambda_e^4 + 6 \sum_{e < f} \lambda_e^2 \lambda_f^2 + 3\mathcal{C}_4(\lambda)$. Since $3s(\lambda)^2 = 3 \sum_e \lambda_e^4 + 6 \sum_{e < f} \lambda_e^2 \lambda_f^2$, subtracting gives $\kappa_4(\lambda) = -2 \sum_e \lambda_e^4 + 3\mathcal{C}_4(\lambda)$, and dividing by 24 yields (8). $\square$

The formulas (7)–(8) express $\mathcal{C}_4$ and Q as polynomials in the entries of any symmetric zero-diagonal matrix; we write Q_m and $\mathcal{C}_{4,m}$ when the matrix is $m \times m$. The following lemma is the main tool for the vertex-peeling argument that controls Q .

Lemma 2.3 (Peeling identity). *Let A be a symmetric zero-diagonal $n \times n$ matrix with entries $A_{ij} = \lambda_{\{i,j\}}$. Write B for the $(n-1) \times (n-1)$ principal submatrix obtained by deleting row and column n , write $x \in \mathbb{R}^{n-1}$ for the last column, and set $M(B) = B^2 - \text{diag}(B^2)$. Then*

$$Q_n(A) = Q_{n-1}(B) + \frac{1}{2} x^\top M(B) x - \frac{1}{12} \sum_{a=1}^{n-1} x_a^4.$$

Proof. The diagonal quartic part splits as

$$\sum_{1 \leq i < j \leq n} A_{ij}^4 = \sum_{1 \leq a < b \leq n-1} B_{ab}^4 + \sum_{a=1}^{n-1} x_a^4.$$

The cycles avoiding vertex n contribute exactly $\mathcal{C}_{4,n-1}(B)$. Since $M(B) = B^2 - \text{diag}(B^2)$, for $a \neq c$ one has $M(B)_{ac} = (B^2)_{ac} = \sum_b B_{ab} B_{bc}$, so $x^\top M(B) x = \sum_{a \neq c} \sum_b x_a B_{ab} B_{bc} x_c$. Because B has zero diagonal, only triples with a, b, c distinct contribute. Each monomial $x_a B_{ab} B_{bc} x_c$ corresponds to the undirected 4-cycle $n-a-b-c-n$, which contributes twice to $x^\top M(B) x$ (from (a, b, c) and (c, b, a)) and eight times to $\mathcal{C}_{4,n}$ (two cyclic orientations, four starting points). Hence the total contribution of all ordered 4-cycles through n is $4x^\top M(B) x$, proving $\mathcal{C}_{4,n}(B, x) = \mathcal{C}_{4,n-1}(B) + 4x^\top M(B) x$. Multiplying by $1/8$ and combining with the diagonal quartic split gives the claimed identity. $\square$

With the cumulant identities and the peeling structure in hand, we bound Q on $\mathcal{D}_{\text{core}}$. The first estimate controls the L^2 norm of e^{4tQ} ; the second shows that e^{4tQ} is close to 1.

Proposition 2.4 (Quartic bounds). *There exist absolute constants $c_1, C_1 > 0$ such that if $d/t \leq c_1$ and $nd^2/t^2 \leq c_1$, then*

$$\int_{\mathcal{D}_{\text{core}}} |e^{4tQ(\lambda)}|^2 e^{-2t\|\lambda\|^2} d\lambda \leq C_1 G_{\text{core}}(d, t),$$

and

$$\int_{\mathcal{D}_{\text{core}}} |e^{4tQ(\lambda)} - 1|^2 e^{-2t\|\lambda\|^2} d\lambda \leq C_1 \frac{d^2}{t^2} G_{\text{core}}(d, t).$$

Proof. Let A be the symmetric zero-diagonal $n \times n$ matrix with entries $A_{ij} = \lambda_{\{i,j\}}$, and split off the last vertex:

$$A = \begin{pmatrix} B & x \\ x^\top & 0 \end{pmatrix}.$$

Define

$$s(B) = \sum_{1 \leq a < b \leq n-1} B_{ab}^2, \quad M(B) = B^2 - \text{diag}(B^2).$$

Since $\|B\|_F^2 = 2s(B)$, submultiplicativity gives $\|M(B)\|_{\text{op}} \leq 2s(B)$ and $\text{tr}(M(B)^2) \leq 4s(B)^2$. On $\mathcal{D}_{\text{core}}$, $s(B) \leq d/t$, so $\|M(B)\|_{\text{op}} \leq 2d/t$ and $\text{tr}(M(B)^2) \leq 4d^2/t^2$. Since Q is real, $|e^{4tQ}|^2 = e^{8tQ}$. Fix $\beta > 0$ and set

$$I_n^{(\beta)} = \int_{\mathcal{D}_{\text{core}}} e^{-2t\|\lambda\|^2} e^{\beta t Q(\lambda)} d\lambda.$$

By Lemma 2.3, $Q_n(A)$ decomposes into $Q_{n-1}(B)$ plus terms quadratic and quartic in x , so for fixed B with $s(B) \leq d/t$ the x -fiber contribution is bounded by

$$\begin{aligned} & e^{-2ts(B)} e^{\beta t Q_{n-1}(B)} \int_{\mathbb{R}^{n-1}} \exp\left(-2t\|x\|^2 + \frac{\beta t}{2} x^\top M(B)x - \frac{\beta t}{12} \sum_a x_a^4\right) dx \\ & \leq e^{-2ts(B)} e^{\beta t Q_{n-1}(B)} \int_{\mathbb{R}^{n-1}} \exp\left(-2t\|x\|^2 + \frac{\beta t}{2} x^\top M(B)x\right) dx. \end{aligned}$$

If d/t is small enough that $\|(\beta/4)M(B)\|_{\text{op}} \leq 1/2$, then $I - (\beta/4)M(B)$ is positive definite, and Lemma 1.11 gives

$$\int_{\mathbb{R}^{n-1}} \exp\left(-2t\|x\|^2 + \frac{\beta t}{2} x^\top M(B)x\right) dx = \left(\frac{\pi}{2t}\right)^{(n-1)/2} \det\left(I - \frac{\beta}{4}M(B)\right)^{-1/2}.$$

Let $u_1, \dots, u_{n-1}$ be the eigenvalues of $(\beta/4)M(B)$. Since $M(B)$ has zero diagonal, $\sum_j u_j = (\beta/4)\text{tr} M(B) = 0$. Also $|u_j| \leq 1/2$, so $-\log(1 - u_j) \leq u_j + Cu_j^2$. Therefore

$$\begin{aligned} \det\left(I - \frac{\beta}{4}M(B)\right)^{-1/2} &= \exp\left(-\frac{1}{2} \sum_j \log(1 - u_j)\right) \\ &\leq \exp\left(C \sum_j u_j^2\right) = \exp\left(\frac{C\beta^2}{16} \text{tr}(M(B)^2)\right) \leq \exp\left(C\beta \frac{d^2}{t^2}\right). \end{aligned}$$

Peeling off vertices one at a time, each step integrates out $n-k$ variables and contributes a Gaussian factor $(\pi/(2t))^{(n-k)/2}$ and a determinant error $\exp(C\beta d^2/t^2)$. Since $\sum_{k=1}^{n-1} (n-k) = d$, the accumulated Gaussian factors give $(\pi/(2t))^{d/2}$, and the n determinant errors multiply to yield $I_n^{(\beta)} \leq \exp(C\beta nd^2/t^2)(\pi/(2t))^{d/2}$. If $nd^2/t^2 \leq c_1$ with c_1 small enough, then $I_n^{(\beta)} \leq C_\beta F(d, t)$. By Lemma 1.12, $F(d, t) \leq C G_{\text{core}}(d, t)$, so taking $\beta = 8$ proves the first estimate.

For the second estimate, $|e^u - 1|^2 \leq u^2(1 + e^{2u})$ with $u = 4tQ(\lambda)$ gives

$$\begin{aligned} \int_{\mathcal{D}_{\text{core}}} |e^{4tQ} - 1|^2 e^{-2t\|\lambda\|^2} d\lambda &\leq 16t^2 \int_{\mathcal{D}_{\text{core}}} Q(\lambda)^2 e^{-2t\|\lambda\|^2} d\lambda \\ &\quad + 16t^2 \int_{\mathcal{D}_{\text{core}}} Q(\lambda)^2 e^{8tQ(\lambda)} e^{-2t\|\lambda\|^2} d\lambda. \end{aligned}$$

For the first term, pass to the centered Gaussian measure γ_t on $\mathbb{R}^d$ with independent coordinates of variance $(4t)^{-1}$, so that $\int_{\mathbb{R}^d} f(\lambda) e^{-2t\|\lambda\|^2} d\lambda = F(d, t) \mathbb{E}_{\gamma_t}[f(g)]$. Write $Q(g) = -\frac{1}{12}D(g) + \frac{1}{8}C(g)$ with $D(g) = \sum_e g_e^4$ and $C(g) = \mathcal{C}_4(g)$. Then $\mathbb{E}_{\gamma_t}[D(g)^2] = O(d^2/t^4)$. For the cross term, every monomial in $D(g)C(g) = \sum_e g_e^4 \sum_\omega \prod_{j=1}^4 g_{e_j(\omega)}$ has at least one coordinate with odd exponent, so $\mathbb{E}_{\gamma_t}[D(g)C(g)] = 0$. For $C(g)^2$, expand into ordered pairs of 4-cycles. A Gaussian moment survives only when every edge variable appears with even total degree, so a fixed ordered cycle has only $O(1)$ partners that contribute. Since there are $O(n^4) = O(d^2)$ ordered cycles and each surviving degree-8

Gaussian moment is $O(t^{-4})$, we get $\mathbb{E}_{\gamma_t}[C(g)^2] = O(d^2/t^4)$. Therefore $\mathbb{E}_{\gamma_t}[Q(g)^2] = O(d^2/t^4)$, and hence

$$\int_{\mathcal{D}_{\text{core}}} Q(\lambda)^2 e^{-2t\|\lambda\|^2} d\lambda \leq C \frac{d^2}{t^4} G_{\text{core}}(d, t).$$

For the second term, Cauchy–Schwarz gives

$$\int_{\mathcal{D}_{\text{core}}} Q(\lambda)^2 e^{8tQ(\lambda)} e^{-2t\|\lambda\|^2} d\lambda \leq \left(\int_{\mathcal{D}_{\text{core}}} Q(\lambda)^4 e^{-2t\|\lambda\|^2} d\lambda \right)^{1/2} \left(\int_{\mathcal{D}_{\text{core}}} e^{16tQ(\lambda)} e^{-2t\|\lambda\|^2} d\lambda \right)^{1/2}.$$

The second factor is $O(G_{\text{core}}(d, t)^{1/2})$ by the already proved exponential estimate with $\beta = 16$. For the first factor, Lemma 1.9 applied to the degree-4 Gaussian polynomial $Q(g)$ with $q = p = 4$ yields

$$\mathbb{E}_{\gamma_t}[Q(g)^4] \leq C \mathbb{E}_{\gamma_t}[Q(g)^2]^2 \leq C \frac{d^4}{t^8}, \quad \text{so} \quad \int_{\mathcal{D}_{\text{core}}} Q(\lambda)^4 e^{-2t\|\lambda\|^2} d\lambda \leq C \frac{d^4}{t^8} G_{\text{core}}(d, t).$$

Combining gives $\int_{\mathcal{D}_{\text{core}}} Q^2 e^{8tQ} e^{-2t\|\lambda\|^2} d\lambda \leq C d^2 t^{-4} G_{\text{core}}(d, t)$. Multiplying by $16t^2$ proves the second estimate. $\square$

The last cumulant-level ingredient bounds the fifth-order phase P .

Lemma 2.5 (Quintic bound). *There is an absolute constant $C'_3 > 0$ such that*

$$\int_{\mathcal{D}_{\text{core}}} P(\lambda)^2 e^{-2t\|\lambda\|^2} d\lambda \leq C'_3 \frac{n^5}{t^5} G_{\text{core}}(d, t).$$

Proof. By definition, $\kappa_5(\lambda) = \mathbb{E}[X_\lambda^5] - 10\mathbb{E}[X_\lambda^3]\mathbb{E}[X_\lambda^2]$. The disconnected even networks in $\mathbb{E}[X_\lambda^5]$ are exactly products of a triangle and a double edge, removed by the subtraction. Thus P is supported on connected even 5-edge networks.

A connected even multigraph with 5 edges has sum of degrees 10 and minimum degree 2, so at most 5 vertices. Choosing at most 5 vertices from $[n]$ gives $O(n^5)$ vertex sets; for each, the number of connected even multigraphs with 5 edges on those vertices is bounded by an absolute constant (distribute 5 among at most $\binom{5}{2} = 10$ edge slots). Therefore P is a universal linear combination of $O(n^5)$ monomials.

Under the Gaussian measure γ_t of variance $(4t)^{-1}$, each monomial in P has degree 5. Fix one such monomial $M(g) = \prod_e g_e^{a_e}$ with $\sum_e a_e = 5$, and let $\mathcal{O}(M) = \{e : a_e \text{ is odd}\}$. Because the underlying multigraph is even, $\mathcal{O}(M)$ is an even simple graph with an odd number of edges; since the total degree is 5, the only possibilities are a triangle or a 5-cycle.

We first describe the two types of monomials M . If $\mathcal{O}(M)$ is a 5-cycle, all five edges carry odd exponent, forcing $a_e = 1$ for each (since $\sum a_e = 5$); the monomial is fully determined by the cycle. If $\mathcal{O}(M)$ is a triangle on $\{i, j, k\}$, three edges carry odd exponent (using at least 3 of the 5 slots), and the remaining two slots form a doubled edge. This doubled edge either coincides with one of the three triangle edges, giving multiplicity pattern $(3, 1, 1)$ on the triangle (3 choices), or joins a triangle vertex to a new vertex ℓ , giving pattern $(1, 1, 1, 2)$ ($O(n)$ choices for ℓ); see Figure 4.

Now we count partners. Let $M'(g) = \prod_e g_e^{b_e}$ be another quintic monomial. If $\mathbb{E}_{\gamma_t}[M(g)M'(g)] \neq 0$, then $a_e + b_e \equiv 0 \pmod{2}$ for every e , so $\mathcal{O}(M') = \mathcal{O}(M)$. For a 5-cycle, M is unique, so the only partner is $M' = M$. For a triangle on $\{i, j, k\}$, any partner M' must share the same odd support and is therefore also of type $(3, 1, 1)$ or $(1, 1, 1, 2)$ on that triangle: $3 + O(n) = O(n)$ partners. Since there are $O(n^5)$ cycle-type monomials and $O(n^4)$ triangle-type monomials, the total number of nonzero pairs is $O(n^5)$.

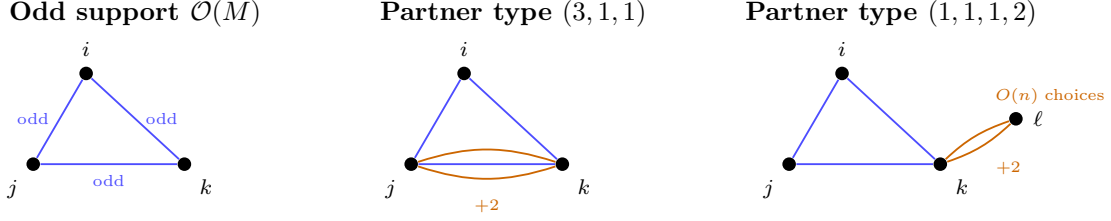


Figure 4: Partner counting when $\mathcal{O}(M)$ is a triangle on $\{i, j, k\}$. The three blue edges have odd exponent (fixed by $\mathcal{O}$); the two orange edge-slots carry the remaining multiplicity. Left: the shared odd support. Middle: the doubled edge lands on a triangle edge (3 choices). Right: the doubled edge goes to a new vertex ℓ ($O(n)$ choices).

Each surviving product $M(g)M'(g) = \prod_e g_e^{m_e}$ has $m_e \in 2\mathbb{N}_0$ and $\sum_e m_e = 10$. Writing $g_e = (4t)^{-1/2}Z_e$ with $Z_e \sim N(0, 1)$, we get $\mathbb{E}_{\gamma_t}[M(g)M'(g)] = (4t)^{-5}\mathbb{E}[\prod_e Z_e^{m_e}] = O(t^{-5})$, since the remaining standard Gaussian moment is bounded by an absolute constant. Multiplying by $F(d, t)$ and using Lemma 1.12 proves the claim. $\square$

2.2 Expansion of the log-characteristic function

The estimates in Section 2 require a factorization of $\psi(\lambda)^{4t}$ into Gaussian, cubic, quartic, quintic, and remainder terms. Lemma 2.6 provides this by expanding $\log \psi(\lambda)$ to fifth order with a sixth-order remainder on $\{s(\lambda) \leq c_2\}$. Lemma 2.7 extracts the Gaussian decay $|\psi(\lambda)|^{4t} \leq e^{-3ts(\lambda)/2}$ on a small ball, which controls both the off-core and residual regions.

Lemma 2.6 (Log-expansion). *There are absolute constants $c_2, C_2 > 0$ such that whenever $s(\lambda) \leq c_2$,*

$$\log \psi(\lambda) = -\frac{1}{2}s(\lambda) - iT(\lambda) + Q(\lambda) + iP(\lambda) + E_6(\lambda)$$

with $|E_6(\lambda)| \leq C_2 s(\lambda)^3$.

Proof. Set $m(u) = \mathbb{E}[e^{iuX_\lambda}]$ and $K(u) = \log m(u)$ for $0 \leq u \leq 1$. By Lemma 1.9 applied to the degree-2 Rademacher polynomial X_λ with $(q, p) = (2, 6)$, one has $\mathbb{E}[|X_\lambda|^6] \leq C \mathbb{E}[X_\lambda^2]^3 = C s(\lambda)^3$. Choose $c_2 \leq 1/2$. If $s(\lambda) \leq c_2$, then Lemma 1.7 applied to $u\lambda$ gives $\operatorname{Re} m(u) = \operatorname{Re} \psi(u\lambda) \in [3/4, 1]$ for $0 \leq u \leq 1$. Hence the principal branch of $\log$ is well-defined along $m([0, 1])$, and $|m(u)| \geq 3/4$ on $[0, 1]$.

Taylor's theorem with integral remainder gives

$$K(1) = \sum_{r=0}^5 \frac{K^{(r)}(0)}{r!} + \frac{1}{5!} \int_0^1 (1-u)^5 K^{(6)}(u) du.$$

Since $m(0) = 1$ and $m'(0) = i\mathbb{E}[X_\lambda] = 0$, $K(0) = K'(0) = 0$. Also $K^{(r)}(0) = i^r \kappa_r(\lambda)$ for $r \geq 2$. For $1 \leq j \leq 6$, one has $m^{(j)}(u) = i^j \mathbb{E}[X_\lambda^j e^{iuX_\lambda}]$ and $|m^{(j)}(u)| \leq \mathbb{E}[|X_\lambda|^j] \leq \mathbb{E}[|X_\lambda|^6]^{j/6}$ by Jensen. The sixth derivative of $K = \log m$ is a finite linear combination of terms

$$m(u)^{-k} \prod_{\ell=1}^k m^{(j_\ell)}(u), \quad j_1 + \cdots + j_k = 6.$$

Since $|m(u)| \geq 3/4$, each factor $m(u)^{-k}$ is $O(1)$, so $|K^{(6)}(u)| \leq C \mathbb{E}[|X_\lambda|^6] \leq C s(\lambda)^3$. Therefore

$$K(1) = \sum_{r=2}^5 \frac{i^r}{r!} \kappa_r(\lambda) + O(s(\lambda)^3).$$

Now $\kappa_2(\lambda) = s(\lambda)$, $\kappa_3(\lambda) = 6T(\lambda)$, $\kappa_4(\lambda)/24 = Q(\lambda)$, and $\kappa_5(\lambda)/120 = P(\lambda)$. Substituting these identities proves the claim. $\square$

On the small ball, the quartic and higher corrections are absorbed by the Gaussian decay.

Lemma 2.7 (Small-ball decay). *There exists an absolute constant $r_0 \in (0, \pi/4)$ such that $|\psi(\lambda)|^{4t} \leq e^{-3t\|\lambda\|^2/2}$ for every integer $t \geq 1$ and every $\lambda \in \mathcal{D}_{r_0}$.*

Proof. By Lemma 2.6, one has $\operatorname{Re} \log \psi(\lambda) = -\frac{1}{2}s(\lambda) + Q(\lambda) + \operatorname{Re} E_6(\lambda)$ whenever $s(\lambda) \leq c_2$. It therefore suffices to show $Q(\lambda) = O(s(\lambda)^2)$. The diagonal quartic term satisfies $\sum_e \lambda_e^4 \leq s(\lambda)^2$. For the ordered 4-cycle term, let A be the symmetric zero-diagonal matrix with entries $A_{ij} = |\lambda_{\{i,j\}}|$. Then

$$|\mathcal{C}_4(\lambda)| \leq \sum_{i_1, i_2, i_3, i_4} A_{i_1 i_2} A_{i_2 i_3} A_{i_3 i_4} A_{i_4 i_1} = \operatorname{tr}(A^4) \leq \operatorname{tr}(A^2)^2 = 4s(\lambda)^2.$$

Hence $|Q(\lambda)| \leq Cs(\lambda)^2$. Combined with $|E_6(\lambda)| \leq Cs(\lambda)^3$ from Lemma 2.6, this gives $\operatorname{Re} \log \psi(\lambda) \leq -\frac{1}{2}s(\lambda) + Cs(\lambda)^2 + Cs(\lambda)^3$ on $\{s(\lambda) \leq c_2\}$. Choose $r_0 \in (0, \pi/4)$ so small that $r_0^2 \leq c_2$, $r_0 \leq 1$, and $Cr_0^2 + Cr_0^4 \leq 1/8$. If $\lambda \in \mathcal{D}_{r_0}$, then $s(\lambda) \leq r_0^2$, so

$$Cs(\lambda)^2 + Cs(\lambda)^3 \leq (Cr_0^2 + Cr_0^4)s(\lambda) \leq \frac{1}{8}s(\lambda),$$

hence $\operatorname{Re} \log \psi(\lambda) \leq -3s(\lambda)/8$ and $|\psi(\lambda)|^{4t} \leq e^{-3ts(\lambda)/2}$. $\square$

2.3 Lower bound on the full core

With the factorization of $\psi(\lambda)^{4t}$ from Lemma 2.6 in hand, we evaluate the core integral by peeling off the higher-order factors one at a time. The cubic phase e^{-4itT} is the only factor that contributes a non-negligible correction: its oscillation reduces the real part by $\binom{n}{3}/(8t)$ (Lemma 2.8). The quartic, quintic, and sixth-order terms are all absorbed into the error via the bounds above. Proposition 2.9 combines these to give the core estimate.

Lemma 2.8 (Cubic phase bound). *For $t \geq n^3$,*

$$\operatorname{Re} K_{\text{core}}(t) = \left(1 - \frac{\binom{n}{3}}{8t} + O\left(\frac{n^6}{t^2} + e^{-cd}\right)\right) F(d, t),$$

where $K_{\text{core}}(t) := \int_{\mathcal{D}_{\text{core}}} e^{-2t\|\lambda\|^2} e^{-4itT(\lambda)} d\lambda$ and $c > 0$ is an absolute constant.

Proof. Because $\mathcal{D}_{\text{core}}$ is centrally symmetric and $T(-\lambda) = -T(\lambda)$, the imaginary part of $K_{\text{core}}(t)$ vanishes. Using $\cos u = 1 - u^2/2 + O(u^4)$:

$$\operatorname{Re} K_{\text{core}} = G_{\text{core}}(d, t) - 8t^2 I_2(t) + O(t^4 I_4(t)),$$

where $I_k(t) := \int_{\mathcal{D}_{\text{core}}} T(\lambda)^k e^{-2t\|\lambda\|^2} d\lambda$. By Lemma 2.1, $T(\lambda) = \sum_\tau \prod_{e \in \tau} \lambda_e$, summed over the $\binom{n}{3}$ triangles of K_n . Since the coordinates are independent under γ_t and each triangle monomial involves three distinct edges, cross terms with distinct triangle supports vanish:

$$\mathbb{E}_{\gamma_t}[T^2] = \binom{n}{3} \mathbb{E}_{\gamma_t}[\lambda_{e_1}^2] \mathbb{E}_{\gamma_t}[\lambda_{e_2}^2] \mathbb{E}_{\gamma_t}[\lambda_{e_3}^2] = \binom{n}{3} (4t)^{-3}.$$

Therefore $8t^2 \mathbb{E}_{\gamma_t}[T^2] F(d, t) = \frac{\binom{n}{3}}{8t} F(d, t)$. Since only diagonal terms survive, and both $\mathcal{D}_{\text{core}}$ and $\mathbb{R}^d \setminus \mathcal{D}_{\text{core}}$ are invariant under coordinate sign flips, the off-diagonal cross terms vanish on each region separately, so the restriction from $\mathbb{R}^d$ to $\mathcal{D}_{\text{core}}$ can be done term by term. For each triangle $\tau = \{e_1, e_2, e_3\}$, Cauchy–Schwarz and Lemma 1.12 give

$$\int_{\mathbb{R}^d \setminus \mathcal{D}_{\text{core}}} \lambda_{e_1}^2 \lambda_{e_2}^2 \lambda_{e_3}^2 e^{-2t\|\lambda\|^2} d\lambda \leq (\mathbb{E}_{\gamma_t}[\lambda_{e_1}^4 \lambda_{e_2}^4 \lambda_{e_3}^4])^{1/2} \cdot O(e^{-c_* d/2}) F(d, t) = O(e^{-cd}) (4t)^{-3} F(d, t),$$

where the Gaussian fourth moments are computed coordinate by coordinate. Summing over $\binom{n}{3}$ triangles: $8t^2 I_2 = \frac{\binom{n}{3}}{8t} F(d, t) + O(e^{-cd}) F(d, t)$. For I_4 , hypercontractivity (Lemma 1.9, $q = 3, p = 4$) gives $\mathbb{E}_{\gamma_t}[T^4] \leq C(\mathbb{E}_{\gamma_t}[T^2])^2 = C\binom{n}{3}^2 (4t)^{-6}$. Since $T^4 \geq 0$, the core integral is bounded by the full Gaussian expectation: $I_4 \leq \mathbb{E}_{\gamma_t}[T^4] F(d, t)$, so $t^4 I_4 \leq Cn^6 t^{-2} F(d, t)$. Substituting into the expansion for $\text{Re } K_{\text{core}}$ and using $G_{\text{core}}(d, t) = F(d, t)(1 + O(e^{-c_* d}))$ (Lemma 1.12) yields the claim. $\square$

The core main term now follows by successively removing Q , P , and E_6 from $\psi(\lambda)^{4t}$ using the bounds above.

Proposition 2.9 (Primary estimate). *Set $K_n := 2^{2d-n+1}(2\pi)^{-d}$. For all sufficiently large n and $t \geq C_0 n^3$,*

$$K_n \text{Re} \int_{\mathcal{D}_{\text{core}}} \psi(\lambda)^{4t} d\lambda = \left(1 - \frac{\binom{n}{3}}{8t} + O\left(\frac{n^2}{t} + \frac{n^{5/2}}{t^{3/2}} + \frac{n^6}{t^2} + e^{-cd}\right)\right) \hat{A}_{n,4t}.$$

Proof. We peel off the factors of $\psi(\lambda)^{4t}$ one at a time, working from the outermost (sixth-order remainder) inward to the cubic phase. Define the intermediate integrals

$$\begin{aligned} M_{\text{core}}(t) &:= \int_{\mathcal{D}_{\text{core}}} e^{-2t\|\lambda\|^2} e^{-4itT(\lambda)} e^{4tQ(\lambda)} e^{4itP(\lambda)} d\lambda, \\ L_{\text{core}}(t) &:= \int_{\mathcal{D}_{\text{core}}} e^{-2t\|\lambda\|^2} e^{-4itT(\lambda)} e^{4tQ(\lambda)} d\lambda. \end{aligned}$$

Since $d \asymp n^2$ and $t \geq C_0 n^3$, we have $d/t \leq c_1$ and $d^3/t^2 \leq c_1$ for a constant $c_1 = c_1(C_0)$ that can be made arbitrarily small by choosing C_0 large. Also $nd^2/t^2 = 2d^3/((n-1)t^2) \leq 2d^3/t^2$, so for all sufficiently large n the hypotheses of Lemma 2.6 and Proposition 2.4 hold on $\mathcal{D}_{\text{core}}$. Hence

$$\psi(\lambda)^{4t} = e^{-2t\|\lambda\|^2} e^{-4itT(\lambda)} e^{4tQ(\lambda)} e^{4itP(\lambda)} e^{4tE_6(\lambda)}$$

throughout $\mathcal{D}_{\text{core}}$, with $|4tE_6(\lambda)| \leq 4C_2 ts(\lambda)^3 \leq Cd^3/t^2$.

We first replace $\psi(\lambda)^{4t}$ by $M_{\text{core}}(t)$, absorbing the sixth-order remainder. If n is large enough that d^3/t^2 is small, then $|e^z - 1| \leq 2|z|$ for $|z| \leq 1$ gives $|e^{4tE_6(\lambda)} - 1| \leq Cts(\lambda)^3$. Therefore

$$\begin{aligned} \left| \text{Re} \left(\int_{\mathcal{D}_{\text{core}}} \psi(\lambda)^{4t} d\lambda \right) - \text{Re } M_{\text{core}}(t) \right| &\leq \left| \int_{\mathcal{D}_{\text{core}}} \psi(\lambda)^{4t} d\lambda - M_{\text{core}}(t) \right| \\ &= \left| \int_{\mathcal{D}_{\text{core}}} e^{-2t\|\lambda\|^2} e^{-4itT(\lambda)} e^{4tQ(\lambda)} e^{4itP(\lambda)} (e^{4tE_6(\lambda)} - 1) d\lambda \right| \\ &\leq \left(\int_{\mathcal{D}_{\text{core}}} |e^{4tQ(\lambda)}|^2 e^{-2t\|\lambda\|^2} d\lambda \right)^{1/2} \left(\int_{\mathcal{D}_{\text{core}}} |e^{4tE_6(\lambda)} - 1|^2 e^{-2t\|\lambda\|^2} d\lambda \right)^{1/2}. \end{aligned}$$

By Proposition 2.4, the first factor is $O(G_{\text{core}}(d, t)^{1/2})$. For the second, $|e^{4tE_6} - 1|^2 \leq Ct^2s(\lambda)^6$, and Lemma 1.10 gives $\int_{\mathcal{D}_{\text{core}}} s(\lambda)^6 e^{-2t\|\lambda\|^2} d\lambda \leq Cd^6 t^{-6} G_{\text{core}}(d, t)$. Combining:

$$\left| \operatorname{Re} \left(\int_{\mathcal{D}_{\text{core}}} \psi(\lambda)^{4t} d\lambda \right) - \operatorname{Re} M_{\text{core}}(t) \right| \leq C \frac{d^3}{t^2} G_{\text{core}}(d, t).$$

Next we strip the quintic phase to pass from $M_{\text{core}}(t)$ to $L_{\text{core}}(t)$. Since $|e^{iu} - 1| \leq |u|$ for real u , one has $|e^{4itP(\lambda)} - 1| \leq 4t|P(\lambda)|$. Cauchy–Schwarz, Proposition 2.4, and Lemma 2.5 yield

$$\begin{aligned} |M_{\text{core}}(t) - L_{\text{core}}(t)| &\leq \left(\int_{\mathcal{D}_{\text{core}}} |e^{4tQ(\lambda)}|^2 e^{-2t\|\lambda\|^2} d\lambda \right)^{1/2} \\ &\quad \times \left(\int_{\mathcal{D}_{\text{core}}} |e^{4itP(\lambda)} - 1|^2 e^{-2t\|\lambda\|^2} d\lambda \right)^{1/2} \\ &\leq CG_{\text{core}}(d, t)^{1/2} \cdot t \left(\int_{\mathcal{D}_{\text{core}}} P(\lambda)^2 e^{-2t\|\lambda\|^2} d\lambda \right)^{1/2} \\ &\leq C \frac{n^{5/2}}{t^{3/2}} G_{\text{core}}(d, t). \end{aligned}$$

Finally we remove the quartic factor to reach $K_{\text{core}}(t)$. By Cauchy–Schwarz and Proposition 2.4,

$$\begin{aligned} |L_{\text{core}}(t) - K_{\text{core}}(t)| &\leq \left(\int_{\mathcal{D}_{\text{core}}} |e^{4tQ(\lambda)} - 1|^2 e^{-2t\|\lambda\|^2} d\lambda \right)^{1/2} \left(\int_{\mathcal{D}_{\text{core}}} e^{-2t\|\lambda\|^2} d\lambda \right)^{1/2} \\ &\leq C \frac{d}{t} G_{\text{core}}(d, t). \end{aligned}$$

It remains to combine the three bounds. By Lemma 2.8, $\operatorname{Re} K_{\text{core}}(t) = (1 - \binom{n}{3}/(8t) + O(n^6/t^2 + e^{-cd}))F(d, t)$. Combining the three peeling bounds with $G_{\text{core}}(d, t) = (1 + O(e^{-c_*d}))F(d, t)$ (Lemma 1.12):

$$\int_{\mathcal{D}_{\text{core}}} \operatorname{Re}(\psi(\lambda)^{4t}) d\lambda = \left(1 - \frac{\binom{n}{3}}{8t} + O\left(\frac{n^2}{t} + \frac{n^{5/2}}{t^{3/2}} + \frac{n^6}{t^2} + e^{-cd} \right) \right) F(d, t).$$

Since $K_n F(d, t) = \hat{A}_{n, 4t}$, the proposition follows. $\square$

3 Off-core estimates

This section shows that the last two terms in (6) (the off-core integral over $\mathcal{B}_\delta \setminus \mathcal{D}_{\text{core}}$ and the residual integral over R_δ) are both $o(\hat{A}_{n, 4t})$ as $n \rightarrow \infty$ with $t/(n^{8/3} \log t) \rightarrow \infty$. The main tool is a Gaussian comparison inequality (Corollary 3.2), which reduces the far-shell bound to a Gaussian characteristic-function estimate. Combined with the core estimate, this yields the expansion in Theorem 1.1 (Section 4).

Fix once and for all $r \in (0, r_0]$, where r_0 is the constant from Lemma 2.7.

3.1 Weak quadratic comparison

Lemma 3.1 compares $\psi(\lambda)$ with its Gaussian counterpart $\psi_G(\lambda)$, with an error controlled by the row-sum influences $I_k(\lambda)$. The proof is the Lindeberg replacement argument from [?, Theorem 3.18], adapted to retain the per-coordinate bound $\sum_m \operatorname{Inf}_m(f)^{3/2}$ rather than passing to the maximum influence; this avoids the normalization $\operatorname{Var}[Q] \leq 1$ assumed there. Corollary 3.2 specializes to the characteristic function setting.

Lemma 3.1 (Lindeberg comparison). *Let $f : [n]^2 \rightarrow \mathbb{R}$ be symmetric with $f(i, i) = 0$, and define the quadratic form and row influences*

$$Q_f(z) := \sum_{1 \leq i, j \leq n} f(i, j) z_i z_j, \quad \text{Inf}_m(f) := \sum_{j=1}^n f(m, j)^2.$$

If $X_1, \dots, X_n$ are independent Rademacher signs, $G_1, \dots, G_n$ are i.i.d. $N(0, 1)$, and $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is C^3 , then

$$|\mathbb{E}[\varphi(Q_f(X))] - \mathbb{E}[\varphi(Q_f(G))]| \leq 6 \|\varphi'''\|_\infty \sum_{m=1}^n \text{Inf}_m(f)^{3/2}.$$

Proof. For $m = 0, 1, \dots, n$, let $Z^{(m)} := (G_1, \dots, G_m, X_{m+1}, \dots, X_n)$ and $F_m := \mathbb{E}[\varphi(Q_f(Z^{(m)}))]$. Then $\mathbb{E}[\varphi(Q_f(X))] - \mathbb{E}[\varphi(Q_f(G))] = \sum_{m=1}^n (F_{m-1} - F_m)$. Fix m . Since f is symmetric and $f(m, m) = 0$, we may write $Q_f(z) = U_m(z) + z_m V_m(z)$, where U_m and V_m depend only on coordinates other than m , and $V_m(z) = 2 \sum_{j \neq m} f(m, j) z_j$. Because $Z^{(m-1)}$ and $Z^{(m)}$ agree on every coordinate except the m -th, evaluating U_m and V_m at either hybrid vector gives the same random variables, independent of both X_m and G_m . Hence $F_{m-1} = \mathbb{E}[\varphi(U_m + X_m V_m)]$ and $F_m = \mathbb{E}[\varphi(U_m + G_m V_m)]$. Applying Taylor's theorem with integral remainder and using that X_m and G_m are centered with variance 1 gives

$$|F_{m-1} - F_m| \leq \frac{\|\varphi'''\|_\infty}{6} (\mathbb{E}[|X_m|^3] + \mathbb{E}[|G_m|^3]) \mathbb{E}[|V_m|^3].$$

Since $\mathbb{E}[|X_m|^3] = 1$ and $\mathbb{E}[|G_m|^3] = 2\sqrt{2/\pi}$, it remains to bound $\mathbb{E}[|V_m|^3]$.

Let $Y_j := G_j$ for $j < m$ and $Y_j := X_j$ for $j > m$. Then $V_m = 2 \sum_{j \neq m} f(m, j) Y_j$, where the Y_j are independent and centered with $\mathbb{E}[Y_j^2] = 1$. Hence $\mathbb{E}[V_m^2] = 4 \text{Inf}_m(f)$. Since each Y_j is either Rademacher or standard Gaussian, $\mathbb{E}[Y_j^4] \leq 3$, so

$$\mathbb{E} \left[\left(\sum_j a_j Y_j \right)^4 \right] = \sum_j a_j^4 \mathbb{E}[Y_j^4] + 6 \sum_{i < j} a_i^2 a_j^2 \mathbb{E}[Y_i^2] \mathbb{E}[Y_j^2] \leq 3 \left(\sum_j a_j^2 \right)^2,$$

Setting $a_j = 2f(m, j)$ gives $\mathbb{E}[V_m^4] \leq 48 \text{Inf}_m(f)^2$. By Cauchy-Schwarz, $\mathbb{E}[|V_m|^3] \leq \mathbb{E}[V_m^2]^{1/2} \mathbb{E}[V_m^4]^{1/2} \leq 8\sqrt{3} \text{Inf}_m(f)^{3/2}$. Therefore

$$|F_{m-1} - F_m| \leq \frac{1 + 2\sqrt{2/\pi}}{6} 8\sqrt{3} \|\varphi'''\|_\infty \text{Inf}_m(f)^{3/2} \leq 6 \|\varphi'''\|_\infty \text{Inf}_m(f)^{3/2}.$$

Summing over m proves the claim. $\square$

Specializing to $\varphi = \cos$ and $\varphi = \sin$ yields the comparison for ψ .

Corollary 3.2 (Gaussian comparison). *Let ψ_G be the characteristic function obtained by replacing the Rademacher entries ξ_i with i.i.d. standard Gaussians g_i . For each row $k \in [n]$, define the row influence $I_k(\lambda) := \sum_{i \neq k} \lambda_{\{i, k\}}^2$, and set $J(\lambda) := \sum_{k=1}^n I_k(\lambda)^{3/2}$. Then*

$$|\psi(\lambda) - \psi_G(\lambda)| \leq \frac{3}{2} J(\lambda) \quad \text{for every } \lambda \in \mathbb{R}^d.$$

Proof. Set $f(i, j) = \lambda_{\{i, j\}}/2$ for $i \neq j$ and $f(i, i) = 0$, so that $Q_f(z) = \sum_{i < j} \lambda_{\{i, j\}} z_i z_j$ and $\text{Inf}_m(f) = \frac{1}{4} I_m(\lambda)$. Applying Lemma 3.1 with $\varphi = \cos$ and $\varphi = \sin$, and using $\|\cos'''\|_\infty, \|\sin'''\|_\infty \leq 1$, gives

$$|\psi(\lambda) - \psi_G(\lambda)| \leq 12 \sum_{m=1}^n \text{Inf}_m(f)^{3/2} = \frac{3}{2} J(\lambda).$$

$\square$

3.2 Pointwise far-shell bounds

The far shell $\mathcal{B}_{\pi/4} \setminus \mathcal{D}_r$ is split into three sub-regions according to the comparison error $J(\lambda)$ and the maximal row-sum $I_{\max}(\lambda)$. On each, we obtain a strict contraction of $|\psi|$.

Proposition 3.3 (Far-shell contraction). *Set $\eta_r := \frac{1}{3}[1 - (1 + 2r^2)^{-1/4}]$ and $I_{\max}(\lambda) := \max_k I_k(\lambda)$. There exist constants $q_{\text{sm}}, q_{\text{big}} \in (0, 1)$ and $a_r > 0$, depending only on r , such that for all $n \geq 2$ and $t \geq 1$ the far shell decomposes as*

$$\begin{aligned}\mathcal{F}_{\text{sm}} &:= \{\lambda \in \mathcal{B}_{\pi/4} \setminus \mathcal{D}_r : J(\lambda) \leq \eta_r\}, \\ \mathcal{F}_{\text{big}} &:= \{\lambda \in \mathcal{B}_{\pi/4} \setminus \mathcal{D}_r : J(\lambda) > \eta_r, I_{\max}(\lambda) \geq 1\}, \\ \mathcal{F}_{\text{mid}} &:= \{\lambda \in \mathcal{B}_{\pi/4} \setminus \mathcal{D}_r : J(\lambda) > \eta_r, I_{\max}(\lambda) < 1\},\end{aligned}$$

and

$$|\psi(\lambda)|^{4t} \leq \begin{cases} q_{\text{sm}}^{4t} & \lambda \in \mathcal{F}_{\text{sm}}, \\ q_{\text{big}}^{4t} & \lambda \in \mathcal{F}_{\text{big}}, \\ \exp(-a_r t n^{-2/3}) & \lambda \in \mathcal{F}_{\text{mid}}. \end{cases}$$

Proof. Set $q_G := (1 + 2r^2)^{-1/4}$, $q_{\text{sm}} := (1 + q_G)/2$, $q_{\text{big}} := ((1 + e^{-8/\pi^2})/2)^{1/2}$, and $a_r := (4/\pi^2)\eta_r^{2/3}$.

Small- J region. Let M be the symmetric matrix with $M_{ij} = \lambda_{\{i,j\}}$ for $i \neq j$ and $M_{ii} = 0$, so that $\sum_{i < j} \lambda_{\{i,j\}} g_i g_j = \frac{1}{2} g^\top M g$. Lemma 1.11 gives

$$\psi_G(\lambda) = \det(I - iM)^{-1/2}, \quad |\psi_G(\lambda)| = \det(I + M^2)^{-1/4}.$$

Because M^2 is positive semidefinite, $\det(I + M^2) \geq 1 + \text{tr}(M^2) = 1 + 2s(\lambda)$, so

$$|\psi_G(\lambda)| \leq (1 + 2s(\lambda))^{-1/4} \leq q_G \quad (\lambda \in \mathcal{B}_{\pi/4} \setminus \mathcal{D}_r).$$

If $\lambda \in \mathcal{F}_{\text{sm}}$, then Corollary 3.2 gives $|\psi(\lambda) - \psi_G(\lambda)| \leq \frac{3}{2}J(\lambda) \leq \frac{3}{2}\eta_r = (1 - q_G)/2$. Therefore $|\psi(\lambda)| \leq q_G + (1 - q_G)/2 = q_{\text{sm}}$, and hence $|\psi(\lambda)|^{4t} \leq q_{\text{sm}}^{4t}$ for $\lambda \in \mathcal{F}_{\text{sm}}$.

Large- $I_{\max}$ region. If $\lambda \in \mathcal{F}_{\text{big}}$, choose k with $I_k(\lambda) = I_{\max}(\lambda)$. By Lemma 1.2,

$$|\psi(\lambda)|^2 \leq \frac{1}{2} + \frac{1}{2} \prod_{i \neq k} \cos(2\lambda_{\{i,k\}}).$$

Since $|\lambda_{\{i,k\}}| \leq \pi/4$, the elementary bound $\cos(2x) \leq 1 - 8x^2/\pi^2 \leq e^{-8x^2/\pi^2}$ for $|x| \leq \pi/4$ implies

$$|\psi(\lambda)|^2 \leq \frac{1}{2} + \frac{1}{2} \exp\left(-\frac{8}{\pi^2} I_k(\lambda)\right) \leq \frac{1}{2} + \frac{1}{2} e^{-8/\pi^2} = q_{\text{big}}^2.$$

Thus $|\psi(\lambda)| \leq q_{\text{big}}$, and hence $|\psi(\lambda)|^{4t} \leq q_{\text{big}}^{4t}$ for $\lambda \in \mathcal{F}_{\text{big}}$.

Middle region. If $\lambda \in \mathcal{F}_{\text{mid}}$, the same bound gives

$$|\psi(\lambda)|^2 \leq \frac{1}{2} + \frac{1}{2} \exp\left(-\frac{8}{\pi^2} I_{\max}(\lambda)\right).$$

Set $x = 8I_{\max}(\lambda)/\pi^2 \in [0, 8/\pi^2) \subset [0, 1]$. The function $f(x) := e^{x/4} + e^{-3x/4}$ satisfies $f'(x) < 0$ on $[0, 1]$, so $f(x) \leq f(0) = 2$. Equivalently, $(1 + e^{-x})/2 \leq e^{-x/4}$, giving $|\psi(\lambda)|^2 \leq e^{-2I_{\max}(\lambda)/\pi^2}$. Since $J(\lambda) \leq nI_{\max}(\lambda)^{3/2}$, the condition $\lambda \in \mathcal{F}_{\text{mid}}$ gives $I_{\max}(\lambda) \geq (\eta_r/n)^{2/3}$. Hence

$$|\psi(\lambda)|^{4t} \leq \exp\left(-\frac{4}{\pi^2} t I_{\max}(\lambda)\right) \leq \exp\left(-a_r t n^{-2/3}\right) \quad (\lambda \in \mathcal{F}_{\text{mid}}).$$

□

Integrating the three pointwise bounds over the far shell and comparing with $\hat{A}_{n,4t}$ yields the following.

Corollary 3.4 (Far-shell integrals). *Let $q_{\text{sm}}, q_{\text{big}}, a_r$ be the constants from Proposition 3.3, and put $E_r(n, t) := q_{\text{sm}}^{4t} + q_{\text{big}}^{4t} + e^{-a_r t n^{-2/3}}$. Then for every $\delta \in (0, \pi/4)$,*

$$\begin{aligned} 2^{2d-n+1} (2\pi)^{-d} \int_{\mathcal{B}_\delta \setminus \mathcal{D}_r} |\psi(\lambda)|^{4t} d\lambda &\leq 2^{-n+1} E_r(n, t), \\ (2\pi)^{-d} \sum_{\lambda \in \Lambda} \int_{\lambda + (\mathcal{B}_{\pi/4} \setminus \mathcal{D}_r)} |\psi(\gamma)|^{4t} d\gamma &\leq 2^{-n+1} E_r(n, t). \end{aligned}$$

In particular, both quantities are $o(\hat{A}_{n,4t})$ as $n \rightarrow \infty$ with $t/(n^{8/3} \log t) \rightarrow \infty$.

Proof. Since $\mathcal{B}_\delta \setminus \mathcal{D}_r \subset \mathcal{B}_{\pi/4} \setminus \mathcal{D}_r$, integrating the three pointwise bounds from Proposition 3.3 over $\mathcal{B}_\delta \setminus \mathcal{D}_r$ gives

$$\int_{\mathcal{B}_\delta \setminus \mathcal{D}_r} |\psi(\lambda)|^{4t} d\lambda \leq \left(\frac{\pi}{2}\right)^d E_r(n, t).$$

Multiplying by $2^{2d-n+1} (2\pi)^{-d} (\pi/2)^d = 2^{-n+1}$ proves the first inequality. For the translated far shell, let $S := \mathcal{B}_{\pi/4} \setminus \mathcal{D}_r$. Then for $\lambda \in \Lambda$ and $\mu \in S \subset \mathcal{B}_{\pi/4}$, Lemma 1.3(i),(iv) gives $|\psi(\lambda + \mu)|^{4t} = |\psi(\mu)|^{4t}$. Therefore

$$(2\pi)^{-d} \sum_{\lambda \in \Lambda} \int_{\lambda + S} |\psi(\gamma)|^{4t} d\gamma = 2^{2d-n+1} (2\pi)^{-d} \int_S |\psi(\mu)|^{4t} d\mu,$$

and the same integrated bound over S yields the second inequality.

It remains to show $2^{-n+1} E_r(n, t) = o(\hat{A}_{n,4t})$ as $n \rightarrow \infty$ with $t/(n^{8/3} \log t) \rightarrow \infty$; all asymptotic notation below refers to this limit. Fix $q \in (0, 1)$. Then

$$\log \frac{2^{-n+1} q^{4t}}{\hat{A}_{n,4t}} = -2d \log 2 + \frac{d}{2} \log(8\pi t) + 4t \log q.$$

Since $d \asymp n^2$ and $t/(n^{8/3} \log t) \rightarrow \infty$, one has $t/(d \log t) \rightarrow \infty$, so the negative linear term dominates and $2^{-n+1} q^{4t} = o(\hat{A}_{n,4t})$. For the exponential term,

$$\log \frac{2^{-n+1} e^{-a_r t n^{-2/3}}}{\hat{A}_{n,4t}} = -2d \log 2 + \frac{d}{2} \log(8\pi t) - a_r t n^{-2/3}.$$

Since $t/(n^{8/3} \log t) \rightarrow \infty$, we have $t n^{-2/3}/(d \log t) \rightarrow \infty$, so the negative linear term dominates and $2^{-n+1} e^{-a_r t n^{-2/3}} = o(\hat{A}_{n,4t})$. Summing the three terms proves the final assertion. $\square$

3.3 Residual estimate

Proposition 3.5 combines the small-ball decay (Lemma 2.7), the integrated far-shell bound (Corollary 3.4), and the odd-cell bound (Lemma 1.6) to show that the off-core and residual integrals are both negligible.

Proposition 3.5 (Residual estimate). *Set $\delta^2 = 2d/t$ and $K_n := 2^{2d-n+1}(2\pi)^{-d}$. Then*

$$K_n \int_{\mathcal{B}_\delta \setminus \mathcal{D}_{\text{core}}} |\psi(\lambda)|^{4t} d\lambda = o(\hat{A}_{n,4t}),$$

$$(2\pi)^{-d} \left| \int_{R_\delta} \psi(\gamma)^{4t} d\gamma \right| = o(\hat{A}_{n,4t}),$$

as $n \rightarrow \infty$ with $t/(n^{8/3} \log t) \rightarrow \infty$. Moreover, for $t \geq C_0 n^3$ and all sufficiently large n , both bounds are at most $C e^{-cn^2} \hat{A}_{n,4t}$ for absolute constants $c, C > 0$.

Proof. All asymptotic notation below is as $n \rightarrow \infty$ with $t/(n^{8/3} \log t) \rightarrow \infty$. Since $d \asymp n^2$, we have $d/t \rightarrow 0$, so eventually $\delta < r$. Also $K_n G_{\text{core}}(d, t) = [1 + o(1)] \hat{A}_{n,4t}$ by the proof of Proposition 2.9.

Outside the core. Set $\mathcal{A}_{\text{loc}}(t, \delta, r) := (\mathcal{B}_\delta \cap \mathcal{D}_r) \setminus \mathcal{D}_{\text{core}}$. If $\lambda \in \mathcal{A}_{\text{loc}}(t, \delta, r)$, then $\|\lambda\|^2 > d/t$, so Lemma 2.7 gives

$$|\psi(\lambda)|^{4t} \leq e^{-3t\|\lambda\|^2/2} \leq e^{-d/2} e^{-t\|\lambda\|^2}.$$

Therefore

$$\int_{\mathcal{A}_{\text{loc}}(t, \delta, r)} |\psi(\lambda)|^{4t} d\lambda \leq e^{-d/2} \left(\frac{\pi}{t}\right)^{d/2} \leq C e^{-c_4 d} G_{\text{core}}(d, t), \quad c_4 = \frac{1}{2} - \frac{\log 2}{2} > 0,$$

and hence $K_n \int_{\mathcal{A}_{\text{loc}}(t, \delta, r)} |\psi(\lambda)|^{4t} d\lambda = o(\hat{A}_{n,4t})$. By Corollary 3.4, $K_n \int_{\mathcal{B}_\delta \setminus \mathcal{D}_r} |\psi(\lambda)|^{4t} d\lambda = o(\hat{A}_{n,4t})$. Since $\mathcal{B}_\delta \setminus \mathcal{D}_{\text{core}} = \mathcal{A}_{\text{loc}}(t, \delta, r) \sqcup (\mathcal{B}_\delta \setminus \mathcal{D}_r)$, this proves the first claim.

Odd residual. By Lemma 1.5, $R_\delta = R_\delta^{\text{even}} \cup R_\delta^{\text{odd}}$; we bound each piece separately. For $\gamma \in R_\delta^{\text{odd}}$, Lemma 1.6 gives $|\psi(\gamma)|^2 \leq \frac{1}{2}$, so $|\psi(\gamma)|^{4t} \leq 2^{-2t}$. Hence

$$(2\pi)^{-d} \int_{R_\delta^{\text{odd}}} |\psi(\gamma)|^{4t} d\gamma \leq 2^{-2t}.$$

As in the proof of Corollary 3.4, $2^{-2t} = o(\hat{A}_{n,4t})$.

Translated near shell. Since $\mathcal{B}_{\pi/4} \setminus \mathcal{B}_\delta \subset (\mathcal{D}_r \setminus \mathcal{B}_\delta) \cup (\mathcal{B}_{\pi/4} \setminus \mathcal{D}_r)$, the even residual is bounded by a translated near-shell term plus a translated far-shell term. For the translated near shell, if $\mu \in \mathcal{D}_r \setminus \mathcal{B}_\delta$, then some coordinate of μ has absolute value at least δ , so $\|\mu\|^2 \geq \delta^2$, and Lemma 2.7 gives $|\psi(\mu)|^{4t} \leq e^{-t\delta^2/2} e^{-t\|\mu\|^2}$. Hence

$$\int_{\mathcal{D}_r \setminus \mathcal{B}_\delta} |\psi(\mu)|^{4t} d\mu \leq e^{-t\delta^2/2} \left(\frac{\pi}{t}\right)^{d/2}.$$

For $\lambda \in \Lambda$ and $\mu \in \mathcal{D}_r \setminus \mathcal{B}_\delta \subset \mathcal{B}_{\pi/4}$, Lemma 1.3(i),(iv) gives $|\psi(\lambda + \mu)|^{4t} = |\psi(\mu)|^{4t}$. Therefore, since $t\delta^2/2 = d$,

$$(2\pi)^{-d} \sum_{\lambda \in \Lambda} \int_{\lambda + (\mathcal{D}_r \setminus \mathcal{B}_\delta)} |\psi(\gamma)|^{4t} d\gamma \leq 2^{2d-n+1} (2\pi)^{-d} e^{-d} \left(\frac{\pi}{t}\right)^{d/2} = e^{-c_* d} \hat{A}_{n,4t},$$

where $c_* = 1 - \frac{1}{2} \log 2$ is the constant from Lemma 1.12.

Translated far shell. Finally, the second inequality of Corollary 3.4 gives $o(\hat{A}_{n,4t})$ for the translated far-shell contribution. Summing the odd, translated near-shell, and translated far-shell bounds proves the second claim.

Exponential bound for $t \geq C_0 n^3$. We verify each bound is $O(e^{-cn^2}) \hat{A}_{n,4t}$. The local annulus and translated near-shell are already expressed as $C e^{-c_4 d} G_{\text{core}}(d, t)$ and $e^{-c_* d} \hat{A}_{n,4t}$, respectively; since $d = \binom{n}{2} \geq n^2/3$, both are $O(e^{-cn^2}) \hat{A}_{n,4t}$.

For the remaining pieces, use the lower bound $\hat{A}_{n,4t} \geq c_0 2^{d-n+1} (2\pi t)^{-d/2}$, which follows from $K_n F(d, t) = (1 + O(e^{-c_* d})) \hat{A}_{n,4t}$.

Corollary 3.4 bounds both far-shell contributions by $2^{-n+1} E_r(n, t)$. Since $q_{\text{sm}}, q_{\text{big}} < 1$ are absolute constants, for n large enough that $4 \log(1/q_{\text{sm}}), 4 \log(1/q_{\text{big}}) \geq a_r n^{-2/3}$ one has $E_r(n, t) \leq 3 \exp(-a_r t n^{-2/3})$, giving

$$\frac{2^{-n+1} E_r}{\hat{A}_{n,4t}} \leq C (\pi t)^{d/2} \exp(-a_r t n^{-2/3}).$$

The exponent $a_r t n^{-2/3} - (d/2) \log(\pi t)$ is increasing for $t \geq d n^{2/3} / (2a_r) \asymp n^{8/3}$, and at $t = C_0 n^3$ it equals $a_r C_0 n^{7/3} - O(n^2 \log n) \geq c n^{7/3}$, so the ratio is $O(e^{-c n^2})$.

The odd residual satisfies $2^{-2t} / \hat{A}_{n,4t} \leq C (2\pi t)^{d/2} 2^{-2t-d+n-1}$. Since $(2t + d - n) \log 2$ grows linearly in t while $(d/2) \log(2\pi t)$ grows logarithmically, the ratio is $O(e^{-c n^3}) \leq O(e^{-c n^2})$ for $t \geq C_0 n^3$. $\square$

4 Proof of main theorem

We now combine the primary estimate (Proposition 2.9) and the off-core bounds (Proposition 3.5) to prove Theorem 1.1.

Proof of Theorem 1.1. Fix n sufficiently large and $t \geq C_0 n^3$. By the three-way decomposition (6),

$$\mathbb{P}(S_{4t} = 0) = K_n \operatorname{Re} \int_{\mathcal{D}_{\text{core}}} \psi^{4t} d\lambda + \operatorname{Re} E,$$

where E collects the off-core and residual contributions. Proposition 2.9 gives

$$K_n \operatorname{Re} \int_{\mathcal{D}_{\text{core}}} \psi(\lambda)^{4t} d\lambda = \left(1 - \frac{\binom{n}{3}}{8t} + O\left(\frac{n^2}{t} + \frac{n^{5/2}}{t^{3/2}} + \frac{n^6}{t^2} + e^{-c n^2} \right) \right) \hat{A}_{n,4t}.$$

Proposition 3.5 gives $|E| = O(e^{-c n^2}) \hat{A}_{n,4t}$, which is absorbed by the existing $O(e^{-c n^2})$ term. Since $N_{n,4t} = 2^{4nt} \mathbb{P}(S_{4t} = 0)$ and $A_{n,4t} = 2^{4nt} \hat{A}_{n,4t}$, dividing by $\hat{A}_{n,4t}$ gives the theorem. $\square$

Corollary 4.1 (Uniform counting). *For every $\varepsilon > 0$ there exists $K > 0$ such that*

$$\left| \frac{N_{n,4t}}{A_{n,4t}} - 1 \right| < \varepsilon \quad \text{for all } n \geq 2 \text{ and } t \geq K n^3.$$

Proof. For $n \geq N_0$ (with N_0 large enough that $C e^{-c N_0^2} < \varepsilon/2$) and $t \geq K n^3$, the expansion gives $|N_{n,4t}/A_{n,4t} - 1| \leq \binom{n}{3}/(8t) + C n^2/t + C n^{5/2}/t^{3/2} + C n^6/t^2 + C e^{-c n^2} \leq C'/K + \varepsilon/2 < \varepsilon$ for K large. For each $n < N_0$, Lemma 1.8 provides T_n such that $|N_{n,4t}/A_{n,4t} - 1| < \varepsilon$ for $t \geq T_n$; enlarging K so that $K n^3 \geq T_n$ for all $n < N_0$ completes the proof. $\square$

Thus $N_{n,4t} \sim A_{n,4t}$ as $t/n^3 \rightarrow \infty$. Conversely, when $t = \Theta n^3$ with $\Theta \geq C_0$ fixed, the error is $O(\Theta^{-2} + 1/n + e^{-c n^2})$, so for Θ large and $n \rightarrow \infty$ the leading correction $1/(48\Theta)$ dominates and $N_{n,4t}/A_{n,4t}$ does not converge to 1: the asymptotics of $N_{n,4t}$ change at the cubic scale.